

Intermittent fasting promotes ILC3s secreting IL-22 contributing to the beigeing of white adipose tissue

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Abstract

Mechanism underlying the metabolic benefit of intermittent fasting remains largely unknown. Here, we reported that intermittent fasting promoted IL-22 production by ILC3s and subsequent beigeing of subcutaneous white adipose tissue. Adoptive transfer of intestinal ILC3s increased beigeing of white adipose tissue in diet-induced-obese mice. Exogenous IL-22 significantly increased the beigeing of subcutaneous white adipose tissue. Deficiency of IL-22 receptor attenuated the beigeing induced by intermittent fasting. Single-cell sequencing of sorted intestinal immune cells revealed that intermittent fasting increased aryl hydrocarbon receptor signaling in ILC3s. Analysis of cell-cell ligand receptor interactions indicated that intermittent fasting may stimulate the interaction of ILC3s with dendritic cells (DCs) and macrophages. These results establish the role of intestinal ILC3s in beigeing of white adipose tissue, suggesting that ILC3/IL-22/IL-22R axis contributes to the metabolic benefit of intermittent fasting.

eLife assessment

This study provides **valuable** findings showing the production of IL-22 from intestinal ILC3 during intermittent fasting promotes beigeing of white adipose tissue. The authors provided **solid** data and mechanistic insight by which IL-22-derived from ILC3 directly induces beigeing.

Introduction

Obesity is defined as an epidemic metabolic disease characterized by excessive fat accumulation as the consequence of long-term energy surplus. Thus, therapeutic management of obesity has been focused on the restoration of energy balance, either by decreasing energy intake or increasing energy expenditure^[1,2]. Among these strategies, intermittent fasting is becoming a popular dietary approach. All intermittent fasting schemes including alternate day fasting, 5:2 intermittent fasting and daily time-restricted feeding have shown the health benefits such as delaying aging^[2,3–6], improving metabolism^[7,8], and enhancing cognition^[9–11]. The mechanism underlying metabolic benefit of intermittent fasting remains largely unknown. Its metabolic benefit was initially attributed to limitation of energy intake. Recent studies have indicated an alternative mechanism involving beiging of white adipose tissue, which accounts for the major plasticity of energy expenditure. Studies by Kim *et al.* have showed that intermittent fasting induces white adipose beiging via stimulation of angiogenesis and macrophage M2 polarization^[12]. On the other hand, studies by Li *et al.*^[13] have indicated that alternate day fasting induces white adipose tissue beiging by shaping the gut microbiota and elevating the fermentation products acetate and lactate. Although gut microbiota is closely related to immune response, it remains unclear whether intestinal immune cells contribute to the metabolic benefit of intermittent fasting.

Innate lymphoid cells (ILCs) are a group of natural immune cells lacking antigen-specific receptors expressed on T cells and B cells^[14]. Based on developmental trajectories, ILCs are divided into five groups: natural killer cells, type 1 ILCs, type 2 ILCs, type 3 ILCs (ILC3s) and lymphoid tissue-inducing cells^[15]. Adipose-Resident type 1 ILCs can promote adipose tissue fibrosis and obesity-associated insulin resistance^[16,17] while type 2 ILCs promote beiging of white adipose tissue and limit obesity^[18–20]. However, the contribution of ILC3s on adipocytes and obesity are less clear. ILC3s can produce interleukin-17 (IL-17) and interleukin-22 (IL-22) in response to extracellular bacteria and fungi^[21]. ILC3s derived IL-22 can enhance the intestinal mucosal barrier function, reduce endotoxemia and inflammation, ameliorate insulin sensitivity^[22,23] and improve the metabolic disorder of polycystic ovary syndrome^[24]. However, ILC3s are also reported to be involved in the induction of obesity^[25], contributing to the metabolic disease^[25–28]. Therefore, the role of ILC3s in metabolic disease seems complex and the role of ILC3s in intermittent fasting and beiging of adipose tissue is not known.

Here, we showed that alternate day fasting promoted the secretion of IL-22 by ILC3s. Further, adoptive transfer of intestinal ILC3s increased thermogenesis in DIO mice. Exogenous IL-22 induced the beiging of white adipose tissue. Deficiency of IL-22 receptor attenuated the beiging of white adipose tissue induced by intermittent fasting. Our study demonstrates that intestinal ILC3-IL-22-IL-22R axis is actively involved in the regulation of adipose tissue beiging. Our findings thus reveal a novel pathway in the dialog between the gut and adipose tissue.

Results

1. Intermittent fasting enhances IL-22 production by intestinal ILC3s

To explore the effect of intermittent fasting on intestinal immune cells, we applied alternate-day fasting to mice fed normal chow diet (NCD-IF group) or high-fat diet (HFD-IF group) (**Fig. 1A**). Intermittent fasting significantly reduced the body weight of mice fed HFD, while demonstrating no effect on food intake rate which was normalized to body weight (**Supplementary Fig. 1A-B**). Moreover, intermittent fasting decreased the respiratory quotient (RQ) on the fasting day while

increased the energy consumption on the feeding day in mice fed HFD (**Supplementary Fig. 1C, D** [↗](#)), improved glucose and lipid metabolism in mice fed HFD (**Supplementary Fig. 1E, 1F** [↗](#)), and promoted white adipose tissue beiging in mice fed NCD or HFD (**Supplementary Fig. 2** [↗](#)).

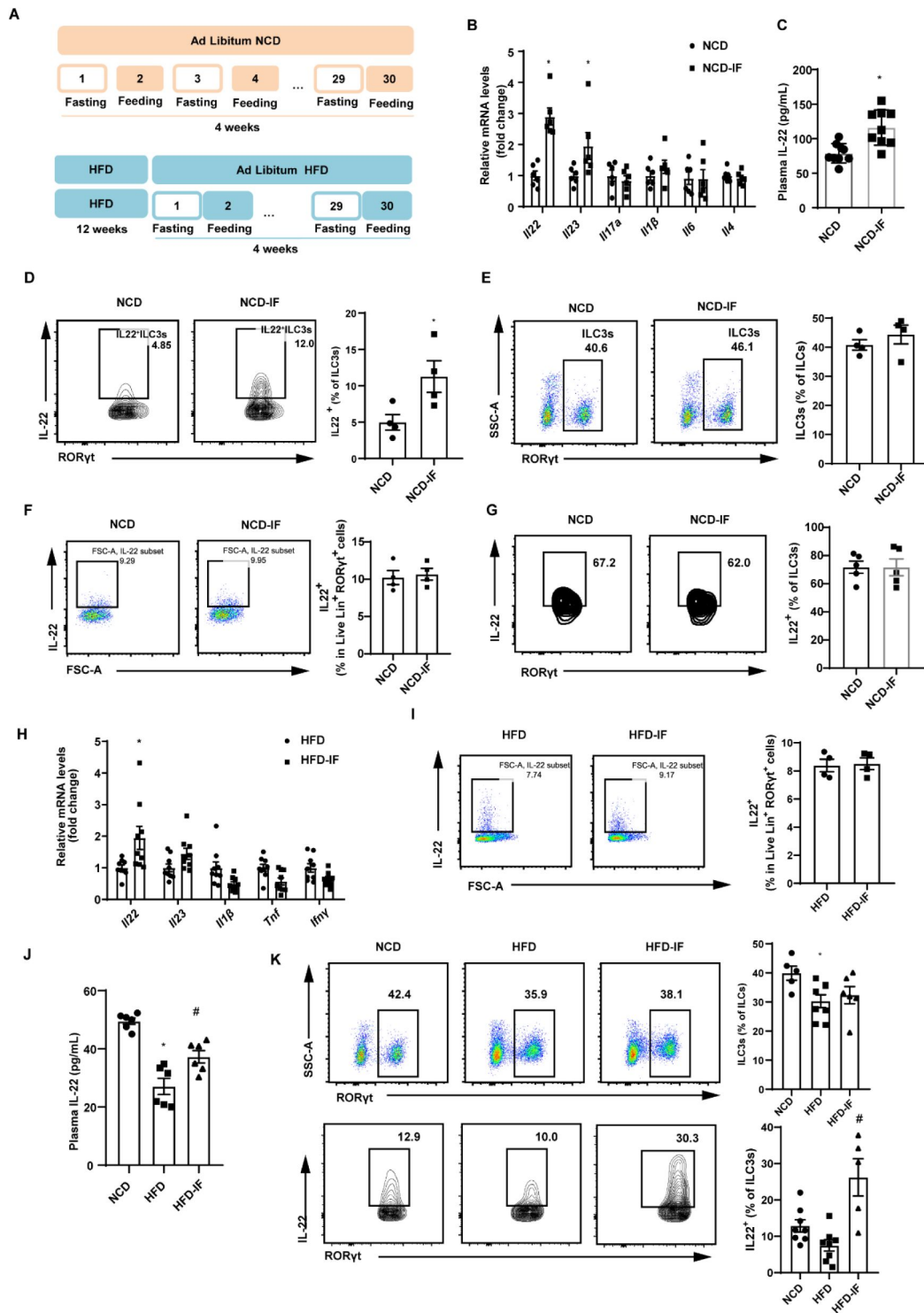


Fig. 1.

Intermittent fasting enhances IL-22 production by intestinal ILC3s.

A Schematic illustration of the alternate day fasting regimen. NCD, normal chow diet. HFD, high-fat diet. We applied alternate-day fasting to mice fed normal chow diet (NCD-IF group) or high-fat diet (HFD-IF group). The control groups were at free access to NCD (NCD group) or HFD (HFD group). n = 9 for each group.

B mRNA expression levels of cytokine genes in the small intestine of NCD and NCD intermittent fasting (NCD-IF) mice. qPCR results were normalized to β -actin. n=6 for each group.

C Protein levels of IL-22 in plasma of NCD and NCD intermittent fasting (NCD-IF) mice. n=9 for each group.

D Flow cytometric analysis of IL-22⁺ cells in live CD127⁺ lineage[−] RORyt⁺ ILC3s from the small intestine lamina propria (siLP) of NCD and NCD intermittent fasting (NCD-IF) mice. Four independent experiments were performed with similar results. n = 4.

E Flow cytometric analysis of RORyt⁺ ILC3s in live CD127⁺ lineage[−] ILCs in the small intestine lamina propria of NCD and NCD intermittent fasting (NCD-IF) mice. n = 4. F Flow cytometric analysis of IL-22⁺ cells in Live Lineage⁺ RORyt⁺ cells in the small intestine lamina propria of mice fed NCD with or without intermittent fasting. n = 4.

G Flow cytometric analysis of IL-22⁺ cells in CD90.2⁺ Lineage[−] RORyt⁺ ILC3s from the SVF cells of sWAT in mice fed NCD with or without intermittent fasting. n= 5.

H mRNA expression levels of cytokine genes in the small intestine of mice fed HFD with or without intermittent fasting. qPCR results were normalized to β -actin. n= 9.

I Flow cytometric analysis of IL-22⁺ cells in Live Lineage⁺ RORyt⁺ cells in the small intestine lamina propria of mice fed HFD with or without intermittent fasting. n=4.

J Levels of IL-22 in plasma of NCD mice, HFD mice and HFD-IF mice. n=6.

K Flow cytometric analysis of RORyt⁺ ILC3s in live CD127⁺ lineage[−] ILCs and flow cytometric analysis of IL-22⁺ cells in live CD127⁺ lineage[−] RORyt⁺ ILC3s from the small intestine lamina propria of NCD mice, HFD mice and HFD-IF mice. n=5-8. * vs NCD, # vs HFD, P < 0.05.

All data represent the mean $\pm$ s.e.m. Statistical significance was determined by unpaired two-tailed Student's *t* test (A-I) or one-way ANOVA with Bonferroni's multiple-comparisons test (J-K). NCD, normal chow diet. HFD, high-fat diet. NCD-IF, normal chow diet with intermittent fasting. HFD-IF, high-fat diet with intermittent fasting.

To explore whether gut immune system contributes to the effects of intermittent fasting on white adipose tissue beigeing, we examined levels of various cytokines in intestine. Notably, mRNA level of IL-22 in the NCD-IF group was significantly higher relevant to the control group (**Fig. 1B**). Consistently, plasma concentration of IL-22 detected by ELISA was also increased (**Fig. 1C**). These results suggest that intermittent fasting increases levels of IL-22 in the intestine and plasma. Since ILC3s are the main source of IL-22 in the small intestine^[29,31], we detected the proportion of ILC3s in the lamina propria of small intestine using flow cytometry. In mice fed NCD, intermittent fasting significantly increased proportion of IL-22 positive ILC3s in the small intestine lamina propria (**Fig. 1D**), while demonstrating no effect on the percentile of total ILC3s (**Fig. 1E**). Interestingly, intermittent fasting did not influence the secretion of IL-22 of T cells marked as Lineage⁺ Ror⁺ cells (**Fig. 1F**). Besides, intermittent fasting didn't alter the levels of ILC3s and IL-22 in mouse adipose tissue (**Fig. 1G** and **Supplementary Fig. 3**). Similar to NCD mice,

intermittent fasting significantly increased mRNA levels of IL-22 in the intestine of HFD mice while didn't influence the secretion of IL-22 by T cells (**Figure 1H-I**). Furthermore, mice fed high-fat diet showed an obvious reduction in the plasma IL-22 (**Fig. 1J**) and percentage of total and IL-22 positive ILC3s (**Fig. 1K**). Further, the decrement of plasma IL-22 (**Fig. 1J**) as well as IL-22 positive ILCs in intestine (**Fig. 1K**) was attenuated by 30 days intermittent fasting.

In order to explain the chronological relationship between ILC3s secreting IL-22 and beiging of white adipose tissue, the mice were exposed to one cycle of intermittent fasting for 2 days. At this time, the body weight of mice didn't change and beige adipocytes haven't been induced (**supplementary Fig 4 A-B**). However, significant increase in proportion of IL-22 positive ILC3s was induced by intermittent fasting for two days whereas total percentile of ILC3s remained unaltered (**supplementary Fig 4 C**). These results indicate that the beiging of white adipose tissue are subsequent to its effect on ILC3s secreting IL-22.

2. Intestinal ILC3s promote beiging of white adipose tissue

Next, we examined whether adoptive transfer of intestinal ILC3s can increase beiging of white adipose tissue in diet-induced obesity (DIO) mice. Intestinal ILC3s were isolated and purified from the small intestinal lamina propria of NCD mice (**Supplementary Fig. 5A**). These cells were defined as lineage⁻CD127⁺KLRG1⁻c-Kit⁺ cells (**Supplementary Fig. 5B**). As shown in **Fig. 2A**, DIO mice transferred with ILC3s demonstrated a significant increment in the proportion of small intestinal ILC3s. Previous researches report that intestinal ILC3s specifically express gut homing receptors CCR7, CCR9, and $\alpha 4\beta 7$ ^[32–34], which may explain transplantation of intestinal ILC3s can migrate mainly to the intestine instead of adipose tissue. Plasma concentration of IL-22 also increased significantly relevant to PBS control (**Fig. 2B**). Relevant to PBS control, adoptive transfer of intestinal ILC3s decreased body weight and weight of sWAT (Subcutaneous adipose tissue) slightly, while had no impact on food intake and the liver weight (**Supplementary Fig 5C-E**). In addition, ILC3s from CD45.1 mouse intestinal lamina propria lymphocytes were adoptively transferred into recipient mice, and CD45.1 positive immune cells were significantly increased in intestine but not in adipose tissue in mice transferred with ILC3s (**Supplementary Fig. 5E**), indicating the feasibility of ILC3s adoptive transfer. Notably, DIO mice transferred with intestinal ILC3s showed improved glucose tolerance (**Fig. 2C**) and decreased levels of random blood glucose (**Fig. 2D**). Cold exposure experiment showed that DIO mice transferred with intestinal ILC3s maintained significantly higher rectal temperatures during a 6-h cold challenge (**Fig. 2E**). Consistently, key thermogenic genes in sWAT including *Ucp1* and *Cidea* were significantly increased (**Fig. 2F**). Size of adipocytes in the sWAT and eWAT was markedly reduced (**Fig. 2G-H**). Furthermore, differentiated SVF cells co-cultured with intestinal ILC3s in vitro (**Fig. 2I**) demonstrated a significant increment in the expression of *Ucp1* and *Pparg* (**Fig. 2J**), as well as a concurrent decrement in the size of lipid droplets (**Fig. 2K**). On the other hand, co-culture with CD127⁻ cells demonstrated no effect (**Fig. 2J-K**). The in vitro experiment indicates the direct effect of ILC3s on adipocytes.

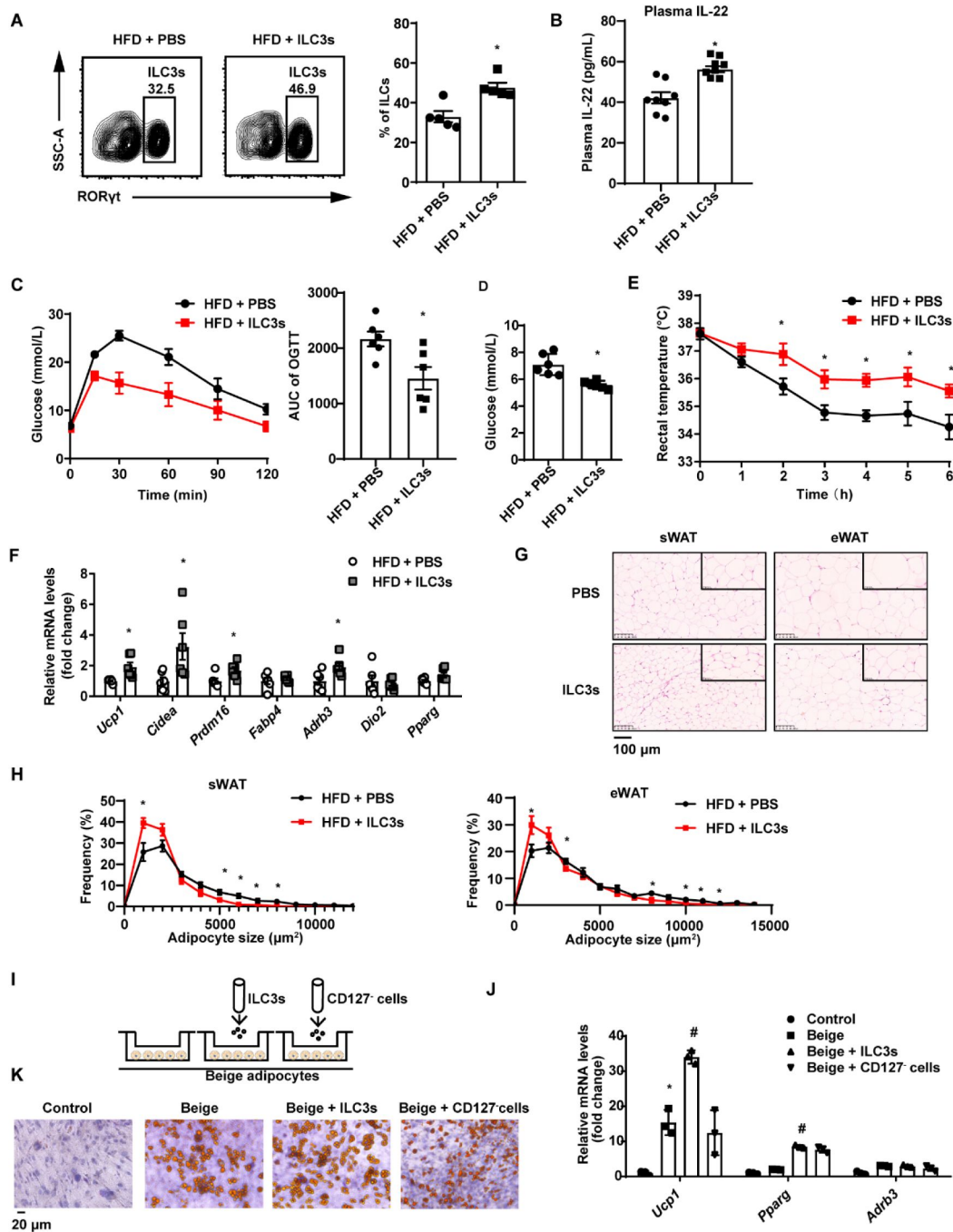


Fig. 2.

ILC3s promote beigeing of white adipose tissue.

6 weeks old male C57BL/6J SPF wild type mice were fed with HFD for 16 weeks and then injected with ILC3s (HFD + ILC3s group) or PBS (HFD + PBS group) intravenously six times in a month. n=8 for each group.

A. Flow cytometric analysis of ROR γ ⁺ ILC3s in live CD127⁺ lineage⁺ ILCs from the siLP of mice transferred with PBS or ILC3s. The proportion of ILC3s in ILCs was shown in the histogram. n=5 for each group.

B. Levels of IL-22 in plasma. n=8 for each group.

C. Oral glucose tolerance test (OGTT) and area under the curve (AUC). n=6 mice/group.

D. Random blood glucose. n=6 mice/group.

E. Rectal temperature of HFD mice transferred with PBS or ILC3s during a 6 h cold challenge (4 °C). n=6 mice/group.

F. qPCR analysis of thermogenic genes in sWAT. n=6 mice/group.

G. Representative images of hematoxylin-and-eosin-stained sections of sWAT, eWAT and BAT from HFD mice transferred with ILC3s or control (n=5).

H. Distribution and average adipocyte size of sWAT and eWAT were shown (n=5).

I. Schematic depicting the coculture of ILC3s with SVF-derived beige adipocytes.

J. qPCR analysis of thermogenic genes in SVF-derived cells. * indicates P<0.05 vs Control, # denotes P<0.05vs Beige. n=3.

K. Oil red O staining of adipocytes after coculture with ILC3s or CD127⁺ cells.

All data represent the means $\pm$ SEM. Statistical significance was determined by unpaired two-tailed Student's *t* test (A-H), one-way ANOVA with Bonferroni's multiple-comparisons test (J).

3. Exogenous IL-22 increases beigeing of white adipose tissue

The role of type 3 immunity in DIO and metabolic syndrome is complex. ILC3-derived IL-22 are beneficial in metabolic syndrome^[22, 35] but can also contribute to metabolic disease^[25–27]. In order to examine the role of promoting beigeing of white adipose tissue of IL-22 in the context of NCD and HFD mice, we intraperitoneally administrated IL-22 at the dose of 4 μ g/kg body weight/every other day for 6 weeks into mice fed NCD or HFD. Saline was used as control. Administration of IL-22 increased its plasma concentration in mice fed either NCD or HFD (**Supplementary Fig. 6A**). Exogenous IL-22 significantly increased oxygen consumption, carbon dioxide production and energy expenditure in mice fed either NCD or HFD (**Fig. 3A-F**). Respiratory quotient (RQ) was significantly reduced in mice fed NCD at dark and mice fed HFD at light(**Fig. 3G-H**). No significant difference was observed for animal activity (**Supplementary Fig. 6B**). In addition, exogenous IL-22 significantly improved glucose tolerance in mice fed HFD (**Fig. 4A**). Interestingly, body weight and food intake were not altered (**Supplementary Fig. 6C-6D**), indicating that the metabolic benefit of IL-22 is not dependent on food intake. Consistent with

previous research, IL-22 administration improves insulin sensitivity without change in body weight^[36]. In addition, IL-22 can increase Akt phosphorylation in muscle, liver and adipose tissues, leading to improvement in insulin sensitivity^[22].

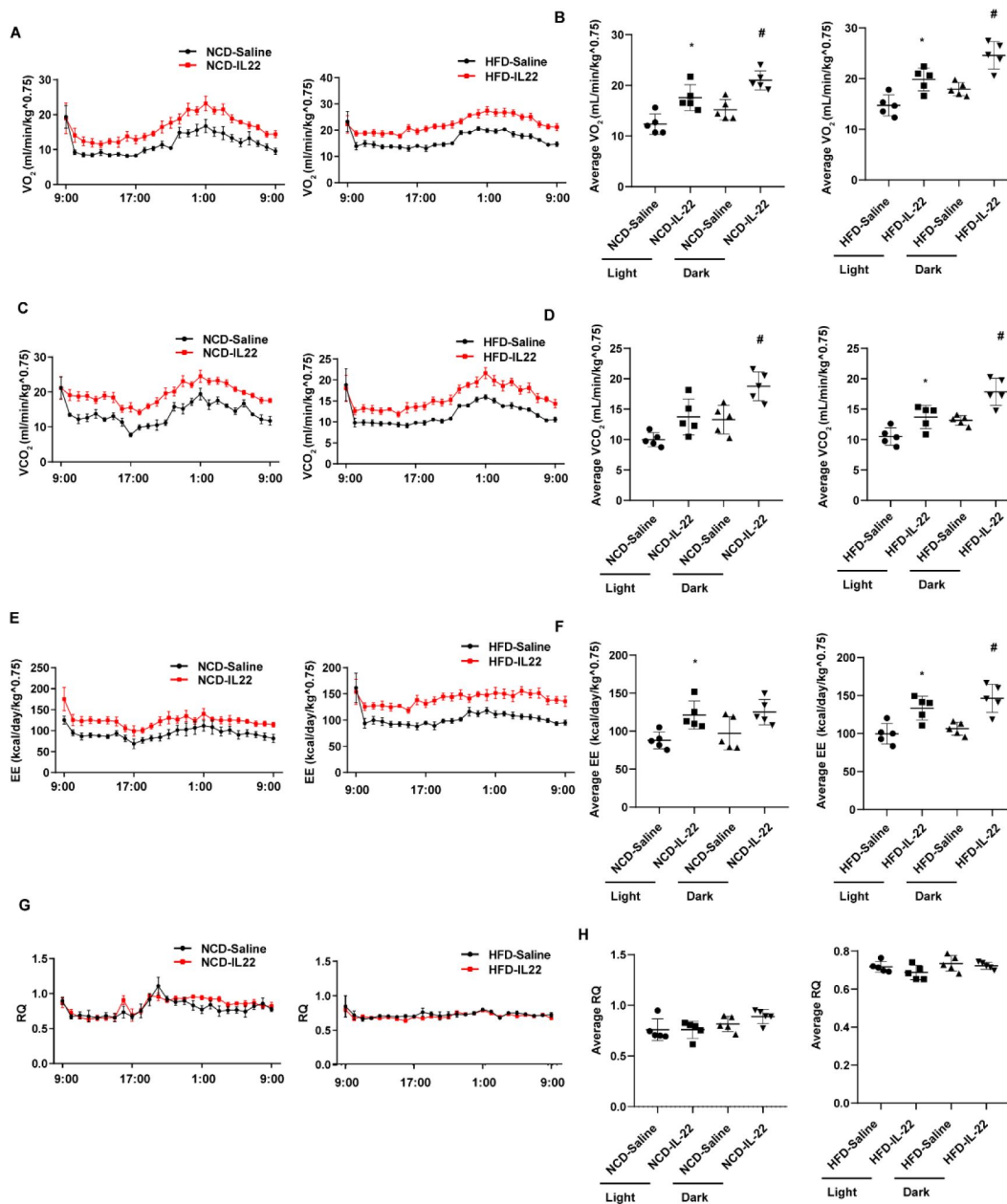


Fig.3.

IL-22 promotes energy expenditure.

6 weeks old male C57BL/6J SPF wild type mice were fed NCD or HFD for 12 weeks and then divided into four groups (NCD-Saline, NCD-IL-22, HFD-Saline, HFD-IL-22). Mice were intraperitoneally injected with 4 μ g/kg IL-22 every other day for six weeks. The control groups were injected with saline.

A VO_2 of mice fed NCD or HFD.

B. Average VO_2 at light and dark respectively.

C. VCO_2 of mice fed NCD or HFD.

D. Average VCO_2 at light and dark respectively.

E. Energy expenditure of mice fed NCD or HFD.

F. Average energy expenditure at light and dark respectively.

G. Respiratory quotient (RQ) of mice fed NCD or HFD

H. Average respiratory quotient at light and dark respectively.

* indicates $P < 0.05$ vs NCD-Saline or HFD-Saline at light. # indicates $P < 0.05$ vs NCD-Saline or HFD-Saline at dark. n = 5.

Statistical significance was determined by one-way ANOVA with Bonferroni's multiple-comparisons test (B, D, F, H).

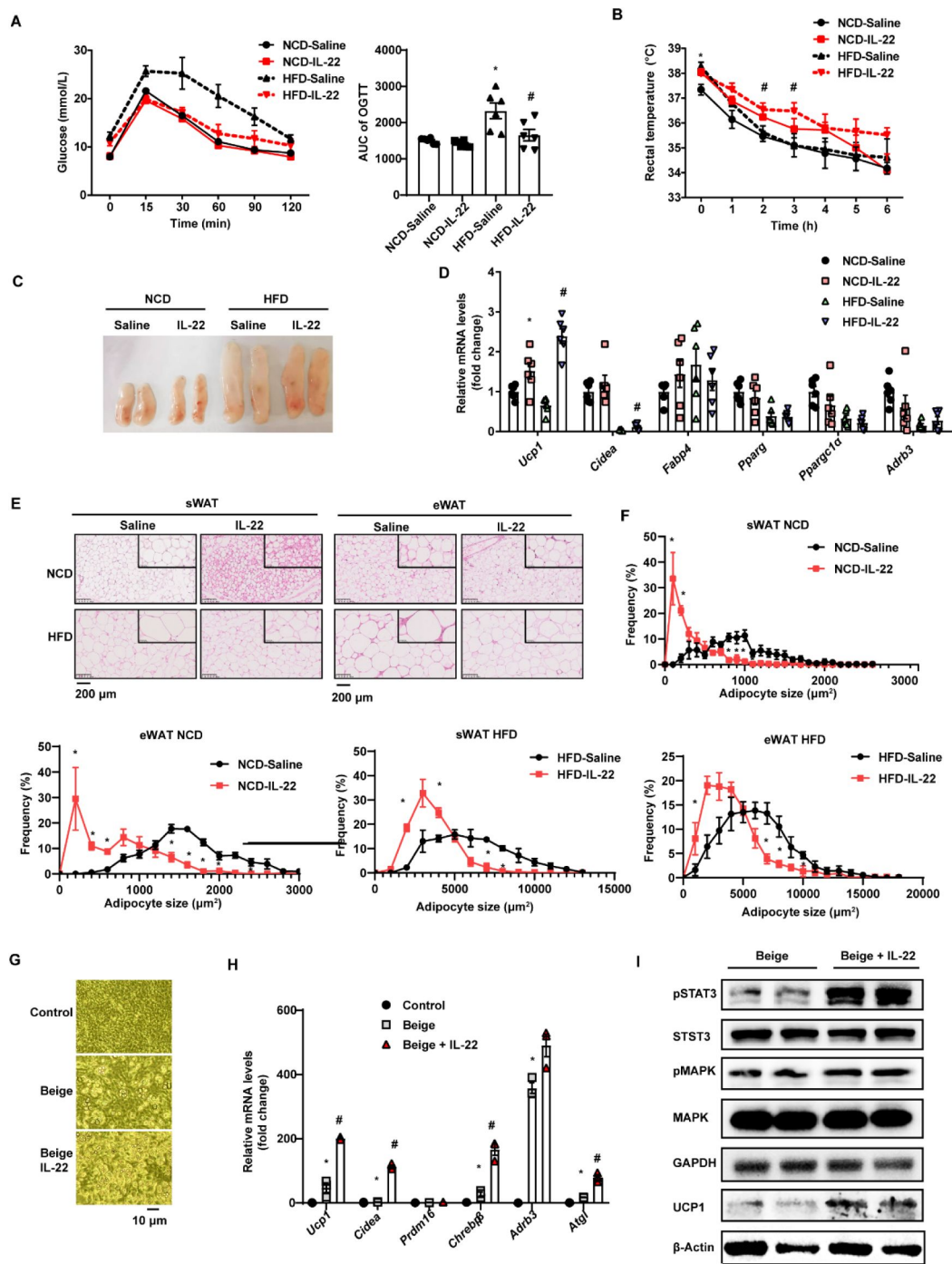


Fig. 4.

IL-22 promotes beigeing of white adipose tissue.

6 weeks old male C57BL/6J SPF wild type mice were fed NCD or HFD for 12 weeks and then divided into four groups (NCD-Saline, NCD-IL-22, HFD-Saline, HFD-IL-22). Mice were intraperitoneally injected with 4 µg/kg IL-22 every other day for 6 weeks. The saline group was injected with saline. n=6 mice/group.

A. Oral glucose tolerance test (OGTT) and area under the curve (AUC). n=6 mice/group.

* indicates $P < 0.05$ vs NCD-Saline; # denotes $P < 0.05$ vs HFD-Saline.

B. Rectal temperature of mice during a 6 h cold challenge (4 °C). n=6. * indicates $P < 0.05$ vs NCD-Saline; # denotes $P < 0.05$ vs HFD-Saline.

C. Representative image of sWAT of the four groups, NCD-Saline, NCD-IL-22, HFD-Saline, HFD-IL-22.

D. qPCR analysis of thermogenic genes of sWAT. n=6 mice/group. * indicates $P < 0.05$ vs NCD-Saline; # denotes $P < 0.05$ vs HFD-Saline.

E. Representative images of hematoxylin-and-eosin-stained sections of sWAT and eWAT (n=5 for each group).

F. The distribution and average adipocyte size of sWAT and eWAT were determined by ImageJ.

G. Phase-contrast microscopic images of SVF cells and adipocytes.

H. qPCR analysis of thermogenic genes in SVF cells and beige adipocytes. n=3. # denotes $P < 0.05$ vs Beige.

I. pSTAT3, STAT3, pMAPK, MAPK, GAPDH, UCP1, β -Actin protein expression in the beige adipocytes or beige adipocytes treated with IL-22 detected by Western blotting. GAPDH and β -Actin was used as the loading control.

All data represent the mean $\pm$ s.e.m. Statistical significance was determined by one-way ANOVA with Bonferroni's multiple-comparisons test (A-D, H) or two-tailed Student's *t* test (F). sWAT, subcutaneous white adipose tissue; eWAT, epididymal white adipose tissue.

We next explored the effect of exogenous IL-22 on thermogenesis induced by 4 °C cold exposure. Exogenous IL-22 rendered mice fed HFD resistant to core temperature drop induced by cold exposure (**Fig. 4B**). Relevant to the pale yellow color in the control animals, subcutaneous fat of obese mice treated with exogenous IL-22 appeared dark yellowish (**Fig. 4C**). The mRNA levels of thermogenic genes such as *Ucp1* were significantly increased by IL-22 (**Fig. 4D**). Adipocyte size of subcutaneous adipose tissue and epididymal adipose tissue was significantly reduced in the animals treated with IL-22 (**Fig. 4E-4F**). These observations suggest that intraperitoneal injection of IL-22 promotes the beigeing of white adipose tissue in mice.

To determine whether IL-22 can directly act on adipocytes to promote their beigeing, adipose tissue SVF was isolated and induced for beige differentiation. IL22ra1 mainly expressed on mature adipocytes (**Supplementary Fig. 6E**). IL-22 at the dose of 100 ng/mL was continuously administered during beige differentiation. IL-22 significantly increased SVF beige differentiation evidenced by cell morphology, increment in mRNA levels of genes relevant to thermogenesis, including *Ucp1* and *Cidea*, and protein level of UCP1 (**Fig. 4G-I**, **Supplementary Fig. 6E**). IL-22 did not alter the levels of genes related to adipogenesis (**Supplementary Fig. 6F**). As expected,

IL-22 increased the phosphorylation of STAT3 and MAPK (**Fig. 4I** [↗](#), **Supplementary Fig. 6G** [↗](#)), which can increase the expression of thermogenic genes^{[37 [↗](#), 38 [↗](#)]}. These results suggest that IL-22 can directly stimulate beiging of white adipose tissue.

4. IL-22R knock-out blocks beiging induced by intermittent fasting

To explore whether IL-22R mediates the effect of intermittent fasting on the beiging of white adipose tissue, IL-22R knockout (IL-22RKO) mice and wild-type (WT) littermates were subjected to alternate-day fasting diet for 30 days. Rectal temperature of WT-IF and IL-22RKO-IF mice was monitored during two consecutive days of intermittent fasting. On the fasting day, the rectal temperature of IL-22RKO-IF mice was significantly lower than that of WT-IF mice at 16:00 time point (**Fig. 5A** [↗](#)). On the feeding day, the rectal temperature of IL-22RKO-IF mice was lower at three time points: 8:00, 12:00 and 20:00 (**Fig. 5B** [↗](#)). These results indicate that IL-22RKO attenuates the thermogenesis induced by intermittent fasting. Knock-out of IL-22R demonstrated no effect on the glucose tolerance and insulin sensitivity in mice fed NCD (**Fig. 5C-D** [↗](#)). However, the weight of subcutaneous adipose tissue in IL-22RKO-IF mice increased significantly (**Fig. 5E** [↗](#)). mRNA levels of the thermogenic gene *Ucp1* decreased substantially, whereas *Fabp4* increased (**Fig. 5F** [↗](#)). Furthermore, knockout of IL-22R significantly attenuated the increment of multilocular lipid droplets and the decrement of adipocyte size in the subcutaneous fat induced by intermittent fasting (**Fig. 5G-H** [↗](#)), indicating a reduction in beiging.

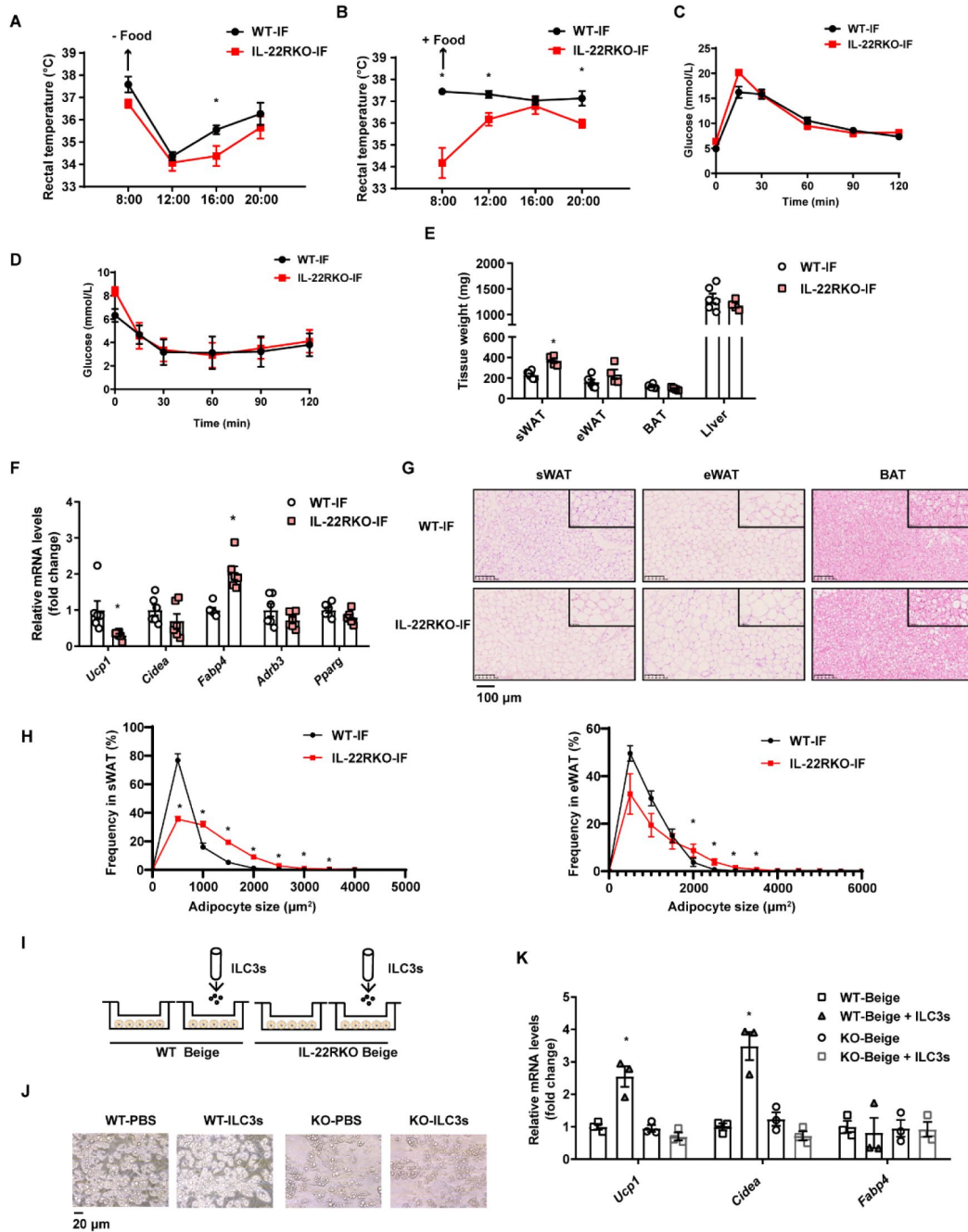


Fig. 5.

IL-22RKO blocks white adipose tissue beigeing induced by intermittent fasting.

8 weeks old IL-22RKO and wild type mice were subjected to alternate day fasting for 30 days. n = 6 mice per group.

A. Rectal temperature of mice at room temperature during the fasting day. n = 6 per group.

B. Rectal temperature of mice at room temperature during the fed day. n = 6 per group.

C. Oral glucose tolerance test (OGTT) of WT-IF mice and IL-22RKO-IF mice. n = 6 per group.

D. Insulin tolerance test (ITT) of WT-IF mice and IL-22RKO-IF mice. n = 6 per group.

E. Tissue weight of sWAT, eWAT, BAT, and liver from WT-IF and IL-22RKO-IF mice. n = 6 mice/group.

F. qPCR analysis of thermogenic genes in sWAT from WT-IF and IL-22RKO-IF mice. n = 6 mice/group.

G. Representative images of hematoxylin-and-eosin-stained sections of sWAT, eWAT and BAT (n = 5 for each group).

H. The distribution and average adipocyte size of sWAT and eWAT were determined by ImageJ.

I. Schematic depicting the co-culture of ILC3s with SVF-induced beige adipocytes from WT or IL-22RKO mice. n = 3. Experiments were repeated 3 times.

J. Phase contrast microscopy images of beige adipocytes differentiated from SVF cells cocultured with or without ILC3s. Shown are representatives from one experiment.

K. qPCR analysis of thermogenic genes in SVF-derived cells co-cultured with or without ILC3s. n = 3.

All data represent the means $\pm$ SEM. Statistical significance was determined by unpaired two-tailed Student's *t* test (A-H) or one-way ANOVA with Bonferroni's multiple-comparisons test (K).

To further explore whether IL-22R mediates the effect of ILC3s on beigeing of adipocytes, intestinal ILC3s isolated from intermittent fasting mice were co-cultured with beige adipocytes from subcutaneous SVF of WT or IL-22RKO mice (**Fig. 5I-J**). Co-culture with ILC3s significantly increased the mRNA levels of thermogenic genes in beige adipocytes derived from WT mice, while demonstrating no effect on beige adipocytes derived from IL-22RKO mice (**Fig. 5K**). These observations indicate that IL-22R mediates the effect of intestinal ILC3s on beigeing of white adipocytes.

5. Profiling of intestinal immune cells in mice fed NCD, HFD or HFD-IF

To explore the mechanism by which intermittent fasting promotes the secretion of IL-22 by ILC3s, live CD45⁺ Lineage (CD3, CD5, B220, CD19, Gr1)[−] cells isolated from the siLP of mice fed NCD, HFD or HFD-IF were subjected to single-cell sequencing (**Supplementary Fig. 7A**). Following quality controls, we analyzed 7455, 4803, 5954 single cells for the NCD, HFD and HFD-IF groups using Seurat-V3.1, respectively. Based on singleR, we identified twenty-five distinct clusters of immune cells (**Fig. 6A** and **Supplementary Fig. 7B**), including ILC1s (Cluster 7 and cluster 16), ILC2s (Cluster 2, cluster 12, cluster 13, and cluster 14), ILC3s (cluster 3 and cluster 5), dendritic cells (Cluster 1, cluster 4, cluster 9, cluster 10, cluster 11, cluster 17, cluster 18, and cluster 21),

eosinophils (Cluster 0 and cluster 6), B cells (Cluster 8 and cluster 24), macrophages (Cluster 15, cluster 20, and cluster 23), NKT cells (Cluster 22) and mast cells (Cluster 19). As expected, ILC3s expressed high RNA levels of *Il7r* (CD127), *Rorc*, and *IL-22* (**Fig. 6B, C** and **Supplementary Fig. 7C, D**). ILC3s contain two distinct cell types with highly similar expression profiles: NCR^+ and NCR^- . These ILC3s were distinguished in our analysis as Cluster 5 and Cluster 3 respectively. NCR^- ILC3s (cluster 3) were CCR6 positive and contained lymphoid tissue inducer (LTi) cells (**Fig. 6B**). Interestingly, NCR^- ILC3s were characterized by high levels of vasoactive intestinal peptide receptor 2 (*Vipr2*) (**Fig. 6C**), which is critical for the migration and function of ILC3s^[29, 39, 40]. The heatmap of cellular composition showed the differences in cell number and percentage among NCD, HFD and HFD IF mice (**Supplementary Fig. 8A**). The change in ILC3 number was consistent with the flow cytometry results. The percentage of IL-22^+ cells in NCR^+ ILC3s was significantly increased by IF in mice fed HFD (**Supplementary Fig. 8B**). Moreover, GSEA revealed that HFD-induced decrement of cytokine–cytokine receptor interaction, in which *IL-22* is included, was reversed by IF (**Supplementary Fig. 8C**). Besides, Neuroactive ligand-receptor interaction, in which *vipr2* is included, was up-regulated in HFD-IF (**Supplementary Fig. 8D**). The mRNA levels of *Vipr2* had a tendency to increase in sorted ILC3s from the small intestine of HFD IF mice compared with that of HFD mice (**Supplementary Fig. 8E**). Further, gene difference analysis revealed that HFD-induced increase of *Zmat4* was significantly attenuated by IF (**Fig. 6D**, **Supplementary Fig. 8F-G**).

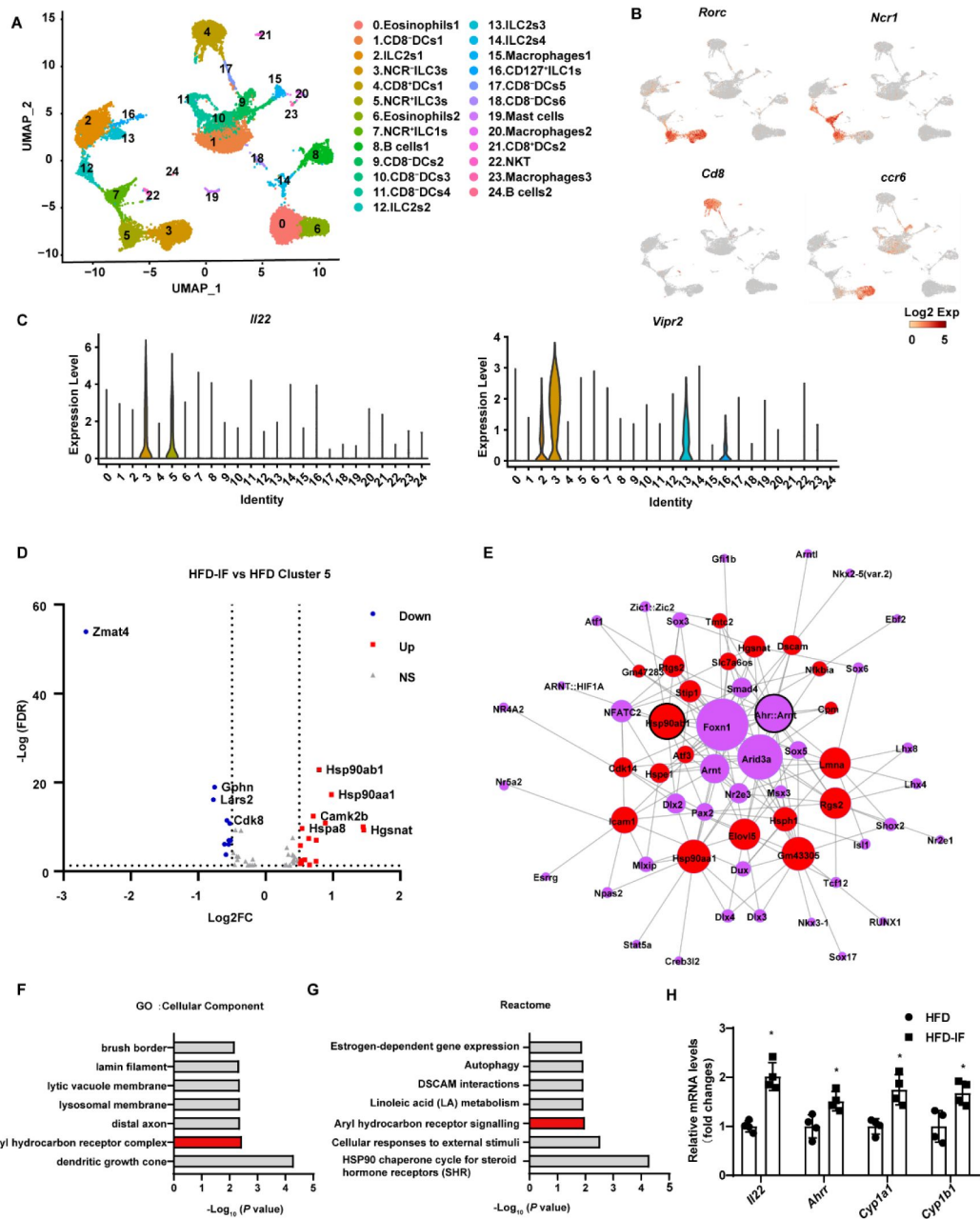


Fig. 6.

Profiling of intestinal immune cells derived from mice fed NCD, HFD and HFD-IF.

Live CD45⁺ Lineage⁻ Cells sorted from mice fed normal chow diet (NCD group), high-fat diet (HFD-group), high-fat diet with alternate day fasting (HFD-IF group) were analyzed using single cell sequencing.

A. Cell subsets in the small intestine lamina propria CD45⁺ Lineage⁻ immune cell atlas. Two-dimensional (2D) representation of cell profiles (dots) from the small intestine lamina propria, colored and numbered by cluster membership.

B-C. UMAP feature plots (B) and violin plots (C) showing RNA expression of cluster markers for the indicated cell populations. UMAP feature plots are based on the UMAP shown in [Figure 6A](#).

D. The volcano plot of differentially expressed genes in Cluster 5 (NCR⁺ ILC3s). The red dots represent upregulated genes in HFD-IF group compared with HFD group, while the blue dots represent downregulated genes in HFD-IF group compared with HFD group. *Hsp90ab1* is one of the notably upregulated genes.

E. Transcription factor prediction using the Jaspar database and TFBS tools. Red dots represent differentially expressed genes, purple dots represent transcription factors, and larger nodes represent more nodes connected to them.

F. Gene ontology (GO) of the top 20 differentially expressed genes in Cluster 5. The cellular component up GO terms in HFD-IF group compared with HFD group.

G. The reactome up terms of Cluster 5 top 20 differential genes in HFD-IF group compared with HFD group.

H. qPCR analysis of AHR target genes in ILC3s sorted from HFD and HFD-IF mice. n=4. *, P<0.05, All data represent the means ± SEM. Statistical significance was determined by unpaired two-tailed Student's *t* test.

Further, the expression level of *Hsp90ab1* in both NCR⁻ ILC3s and NCR⁺ was significantly increased by IF in mice fed HFD ([Fig. 6D](#), [Supplementary 10H](#)). Transcription factor analysis using the JASPAR database and TFBS Tools revealed that the expression of *Hsp90ab1* may be regulated by aryl hydrocarbon receptor (AHR) ([Fig. 6D](#)). AHR is located in the Hsp90: XAP2:p23:Src chaperone protein complex in the cytoplasm without stimulation. Upon binding with the ligand, AhR translocates to the nucleus and heterodimerizes with AhR nuclear translocator (ARNT) to control the transcription of target genes, including AhR repressor (*Ahrr*), *Cyp1a1*, *Cyp1b1* and *Il22*^[41]. Consistently, Gene Ontology analysis of the top 20 increased difference genes revealed that aryl hydrocarbon receptor signaling was one of the key pathways affected by intermittent fasting in HFD mice ([Fig. 6E-F](#)). Furthermore, the AHR target genes, including *Il22*, *Ahrr*, *Cyp1a1*, *Cyp1b1*, increased significantly in ILC3s sorted from HFD-IF mice compared with HFD mice. These results indicates that intermittent fasting increases the production of IL-22 likely via activation of AhR.

6. IF ameliorates the impaired interaction between intestinal myeloid cells and ILC3s

Because dendritic cells (DCs) can influence the production of IL-22 by ILC3s, we next analyzed the change in DCs induced by intermittent fasting. IF significantly attenuated the increment of inflammatory factors, such as *Ccl4*, *Ccl17* and *Ccl22*, of DCs in mice fed HFD ([Fig. 7A](#)). Kyoto Encyclopedia of Genes and Genomes (KEGG) analysis of the top 20 differentially expressed genes in cluster 17 revealed that the increase of inflammation-related pathways induced by HFD was

obviously attenuated by IF (**Fig. 7B-C** [↗](#)). These observations indicate that intermittent fasting may ameliorate the inflammatory state of the intestine by decreasing the NOD-like receptor signaling pathway in DCs (**Fig. 7B-C** [↗](#)).

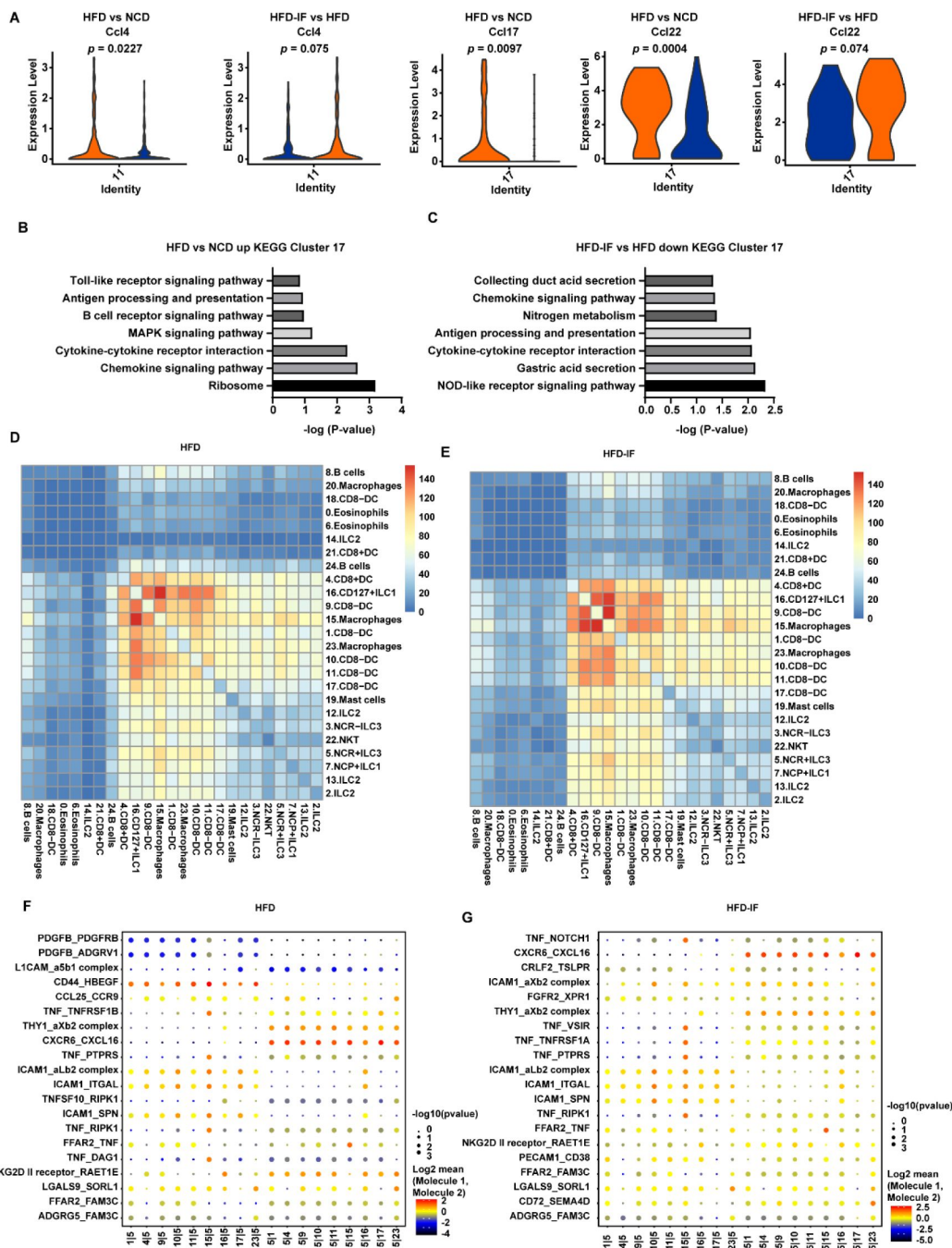


Fig. 7.

Interaction between myeloid cells and ILC3s.

- A. Violin plots showing the RNA expression of chemokines in CD8⁺DCs (cluster 11 and cluster 17).
- B. KEGG enrichment in upregulated genes of CD8⁺DCs of HFD group compared with NCD group (cluster 17).
- C. KEGG enrichment of downregulated genes of CD8⁺DCs of HFD-IF group compared with HFD group (cluster 17).
- D. Heatmap showing the number of significant interactions identified between cell types in sorted siLP immune cells of HFD mice as determined by Cellphone DB. The color represents the number of interactions between cell types, a higher number of interactions (red) and a lower number of interactions (blue).
- E. Heatmap showing the number of significant interactions identified between cell types in sorted siLP immune cells of HFD-IF mice as determined by Cellphone DB. The color represents the number of interactions between cell types: a higher number of interactions (red) and a lower number of interactions (blue).
- F. Interaction pattern of the top 20 protein pairs and the top 20 cell types in sorted siLP immune cells of HFD group mice. The x-axis is the cell type interaction, and the y-axis is the protein interaction. The larger the point is, the smaller the p value. The color represents the average expression, and red to black indicates the level from high to low.
- G. Interaction pattern of the top 20 protein pairs and the top 20 cell types in sorted siLP immune cells of HFD-IF mice. The x-axis is the cell type interaction, and the y-axis is the protein interaction. The larger the point is, the smaller the p value. The color represents the average expression, and red to black indicates the level from high to low.

We next analyzed the intestinal cellular communication networks using CellPhone DB analysis based on homologous gene transformation on all immune cells acquired. CellPhone DB ligand–receptor analysis revealed hundreds of immune-to-immune interactions (**Supplementary Fig. 9A**). Connectome web analysis of siLP immune cells revealed strong interactions among CD8⁺ dendritic cells (DC) (Cluster 4), CD127⁺ ILC1s (Cluster 16), DC8⁺ DC-1 (Cluster 9), Macrophages-1 (Cluster 15), CD8⁺ DC-2 (Cluster 1), Macrophages-2 (Cluster 23), CD8⁺ DC-3 (Cluster 10), CD8⁺ DC-4 (Cluster 11), and CD8⁺ DC-5 (Cluster 17). Notably, NCR⁺ ILC3s strongly interacted with CD8⁺ dendritic cells (DC) (Cluster 4), CD127⁺ ILC1s (Cluster 16), DC8⁺ DC (Cluster 9), macrophages (Cluster 15), CD8⁺ DC (Cluster 1), macrophages (Cluster 23), CD8⁺ DC (Cluster 10), CD8⁺ DC (Cluster 11), and CD8⁺ DC (Cluster 17). ILC subsets, DCs and macrophages were in the central communication hubs of the healthy small intestine (**Supplementary Fig. 9B–C**). Analysis of highly expressed interactions uncovered various uncharacterized and validated signaling pathways implicated in intestine homeostasis in mice (**Supplementary Fig. 9D**). Analysis of the NCD mouse interactome suggests that macrophages (cluster 15) may interact with NCR⁺ ILC3s through CD74_COPA, CD74_MIF, CD44_HBEGF, CD44_FGFR2, CCL4_SLC7A1, and IL1B_ADRB2. Similarly, DCs (cluster 9) may interact with NCR⁺ ILC3s through CD74_COPA, CD74_MIF, IL1B_ADRB2, CD44_HBEGF, and CD44_FGFR2. All these observations indicates that macrophages and DCs play critical roles in the recruitment and maintenance of ILC3s.

HFD significantly reduced the interaction between ILC subsets, DCs and macrophages in the small intestine (**Fig. 7D** and **Supplementary Fig. 9B**). This reduction was reversed by IF (**Fig. 7E** and **Supplementary Fig. 11B**). Analysis of the interactomes in mice fed HFD suggested that IF significantly altered the interacting proteins (**Fig. 7F–G**). To further test this observation, we sorted macrophages and ILC3s from HFD and HFD-IF mice (**Supplementary Fig. 10A**). As shown in **Supplementary Fig. 10B–C**, *Ccl4* and *Cd74* increased significantly in macrophages, while the

receptors expressed on ILC3s remained largely unchanged. Together, these data suggest that IF may promote the production of IL-22 from ILC3s by altering the interactome in intestinal myeloid cells and ILC3s.

Discussion

Our present study demonstrates that intestinal ILC3s are critical for the beigeing of white adipose tissue induced by intermittent fasting. This conclusion is supported by following observations. Firstly, intermittent fasting stimulates the secretion of IL-22 by intestinal ILC3s in either lean mice fed normal chow diet, obese mice fed high-fat diet mice, or mice with metabolic dysfunction induced by HFCD diet. Secondly, a comprehensive set of in vivo and in vitro experiments shows that intermittent fasting promotes adipose tissue beigeing through the intestinal ILC3-IL-22-IL-22R axis.

The role of ILC3s in metabolism remains largely unknown. Our studies provide evidence supporting the metabolic benefit of intestinal ILC3s in intermittent fasting. Adoptive transfer of intestinal ILC3s isolated from lean mice was sufficient to improve the metabolic dysfunctions in DIO mice, including increase of white adipose tissue beigeing and glucose tolerance. Interestingly, adoptive transfer of ILC3s significantly increased their number only in intestine. Further, alternate day fasting did not alter the ILC3s in adipose tissue. These observations indicate that ILC3s in intestine rather than in adipose tissue account for the metabolic benefit. In addition to directly promoting the beigeing of white adipose tissue through IL-22, intestinal ILC3s may enhance the intestinal mucosal barrier^[42], reducing serum LPS and peptidoglycan, thus reducing the inhibitory effect of LPS and peptidoglycan on the beigeing of white adipose tissue^[43]. O'Sullivan et al.^[16] have reported that ILC3s are negligible in white adipose tissue of either lean or obese mice. Similarly, Sasaki *et al.*^[25] have demonstrated that transplanting bone marrow cells from Rag2^{-/-} mice (lack of T cells, B cells) into Il2rg^{-/-} Rag2^{-/-} mice (lack of T cells, B cells, ILC cells) could not increase the number of ILC3s in the adipose tissue of recipient mice. However, it is worth of noting that ILC3s have been detected in adipose tissue by other report. Using the marker Lineage⁻KLRG1⁻IL-7ra⁺Thy-1⁺, ILC3s cells have been reported to account for approximately 20% of lineage⁻KLRG1⁻IL-7ra⁺ cells in adipose tissue. In our study, we also detected ILC3s defined as CD127⁺Lin⁻RORyt⁺ in adipose tissue. These ILC3s accounted for approximately 10% of CD127⁺Lin⁻ cells. ILC3s have also been shown to be present in human adipose tissue^[44]. Importantly, the proportion and density of ILC3s in white adipose tissue of people with obesity increase relevant to healthy people, and positively correlate with BMI in obese patients. Thus, the existence and function of ILC3s in mouse and human adipose tissue deserve further investigation.

Communication between intestine and metabolic organs is currently under active investigation. Our studies identify ILC3s-IL-22-IL-22R as a novel pathway mediating the crosstalk between intestine and adipose tissue. Evidence supporting this conclusion are five folds. (1) Intermittent fasting reverses the reduction of intestinal ILC3s and IL-22 in DIO mice. (2) Adoptive transfer of intestinal ILC3s enhances beigeing of white adipose tissue. (3) Co-culture of intestinal ILC3s with SVF cells increases their differentiation into beige cells. (4) Exogenous IL-22 mimics the effect of intermittent fasting on beigeing of white adipose tissue. (5) Deficiency of IL-22R blocks the IF-induced beigeing of white adipose tissue. In lines with our observation, a series of recent studies have suggested that cytokines can act directly on adipocytes to regulate thermogenesis. For example, $\gamma\delta$ T cells regulate heat production by secreting IL-17^[45]. Deficiency of IL-10 increases energy consumption and renders mice resistant to diet-induced obesity³⁴. IL-27-IL-27Ra signaling promotes thermogenesis, prevents diet-induced obesity, and improves insulin resistance^[46]. IL-33 induces the beigeing of white adipose tissue by activating ILC2s^[18]. Our studies extend the effect of cytokines on thermogenesis to IL-22. IL-22 promotes heat production, renders mice resistant to reduction of body temperature induced by cold exposure, and reduces obesity induced by high-fat diet. Together with previous report showing that IL-22 increases the lipolysis of

adipocytes, our studies suggest that IL-22 can directly act on adipocytes to alter the adipose tissue homeostasis. IL-22 activates IL-22 receptor, which then regulates the expression of downstream inflammatory factors, tissue repair molecules, chemokines, antimicrobial peptides and other molecules via Jak-1 and Tyk-2 dependent phosphorylation of STAT3. Our studies suggest that IL-22 may promote the expression of downstream thermogenic genes in adipose tissue through IL-22R. Deficiency of IL-22R blocks the up-regulation of thermogenic genes induced by intermittent fasting. SVF cells lacking IL-22R demonstrate no response to the up-regulation of thermogenic genes induced by intestinal ILC3. Thus, our results provide novel evidence supporting the concept that intestinal ILC3s modulate beigeing of adipose tissue via IL-22-IL-22R pathway.

The physiological mechanism underlying the secretion of IL-22 by ILC3s remains largely unknown. Our studies reveal the distinct secretion pattern of IL-22 induced by intermittent fasting. Consistently, previous studies have shown that secretion of IL-22 by ILC3s changes between active and quiescent periods throughout the day which is regulated by the cycle patterns of food intake^[47]. These suggest that secretion of IL-22 by ILC3s is altered by feeding rhythm. It is currently unclear how feeding influences the physiological function of ILC3s. Previous studies indicate a mechanism involving neuropeptide vasoactive intestinal peptide (VIP), which is able to stimulate the secretion of IL-22 by ILC3s^[29]. Consistently, our single cell RNAseq data also showed that VIP receptor 2 (VIPR2) is highly expressed in intestinal ILC3s and intermittent fasting activates VIPR2 signaling pathway. Since release of VIP is stimulated by food intake, these observations indicate that VIP-VIPR signaling may coordinate with food intake to drive the production of IL-22. However, conflicting result exists. Studies by Talbot *et al* have shown that VIP inhibits ILC3 secretion of IL-22^[39]. Reasons accounting for this difference remains unknown but may be context dependent. Alternatively, we found that intermittent fasting may promote ILC3s secreting IL-22 through activating AhR signaling. In addition, feeding may decrease the levels of antimicrobial peptides secreted by epithelial cells, while increasing the expression of lipid-binding proteins^[39]. Moreover, segmented filamentous bacteria may account for the effect of feeding on the secretion of IL-22 by intestinal ILC3s by periodically attaching to the epithelial surface^[47]. These alterations may substantially influence the immune networks in intestine, leading to subsequent change in the secretion of IL-22 by ILC3s. In support of this concept, our single cell RNAseq analysis shows a significant change in the intestinal immune cell network involving dendritic cells, macrophages and ILC3s. Further examination should focus on dissecting the novel molecular mechanism by which intestinal dendritic cells and macrophages interact with ILC3s and its consequence on the IL-22 production.

In conclusion, our studies suggest that intermittent fasting can promote the secretion of IL-22 by intestinal ILC3s. IL-22 promotes beigeing of white adipose tissue through IL-22R. Intestinal ILC3s thus may serve as a potential target for the intervention of metabolic disorders.

Materials and Methods

1. Animals

Four-week-old male C57BL/6 and CD45.1 mice were obtained from Charles River Laboratories (Peking, China). IL-22 receptor knockout mice (IL-22R KO mice, KO-00115) were purchased from BRL Medicine Inc. Mice were handled in accordance with the Guide for the Care and Use of Laboratory Animals prepared by the National Academy of Sciences and published by the National Institutes of Health (NIH publication 86-23, revised 1985). Mice were housed in standard rodent cages and maintained in a regulated environment (21-24°C, humidity at 40-70%, 12:12 h light : dark cycle with lights on at 8:00 AM). A normal chow diet (D12450H; Research Diets) and water were available *ad libitum*. Obese mice (DIO) were induced with a high-fat diet (60% fat, D12492; Research Diets) for 12 weeks. Eight-week-old C57BL/6 mice received intraperitoneal injection of

saline control or IL-22 (R&D, 582-ML) at a dose of 4 µg/kg/day every other day for 6 weeks. All experimental procedures were approved by the Animal Care and Use Committee of Peking University.

2. Cell preparation

For isolation of small intestine lamina propria cells, small intestines from euthanized mice were emptied of the contents, excised of Peyer's patches, opened longitudinally and cut into 1 cm pieces. The intraepithelial lymphocytes (IELs) were dissociated from the intestine fragments by first shaking the fragments for 20 min at 37°C in PBS containing 0.3% BSA, 5 mM EDTA and 1 mM dithiothreitol. Vortex the fragments three times with PBS containing 2 mM EDTA. To isolate lamina propria cells, the remaining fragments were minced and digested at 37°C for 50 min in RPMI 1640 medium containing 0.04 mg/mL collagenase IV (Sigma), 0.1 mg/mL deoxyribonuclease (DNase) I (Roche), and 0.5 mg/mL dispase. The digestion suspension was then filtered through a 40 µm cell strainer and centrifuged at 540 g for 6 min. Cell pellets were resuspended in PBS containing 2% FBS for further analysis.

3. Flow cytometry

Single-cell suspensions were preincubated with anti-CD16/32 (clone 2.4G2) for 10 min to block the surface Fc receptors. Then, cell-surface molecules were stained with different antibody combinations for 30 min in cell staining buffer. Dead cells were excluded with Fixable Viability Dye eFluor™ 606 (Invitrogen). For intracellular transcription factor staining, the cells were fixed and permeabilized with a Foxp3 staining buffer set (eBioscience) according to the manufacturer's protocol. Transcription factor staining usually lasted for more than 4 hours at 4°C. The gate strategy for ILC3s was live lineage (CD3, CD5, CD19, B220 and Gr-1)[−] CD127⁺RORγt⁺. For intracellular cytokine staining, the digested cells were first incubated in RPMI 1640 with 10% FBS and 10 ng/ml recombinant murine IL-7 (PeproTech), and stimulated with 50 ng/ml phorbol 12-myristate 13-acetate, 750 ng/ml ionomycin for 3 h and added with 2 µM monensin for the last 2.5 h.

Flow cytometry analyses were performed on an LSR Fortessa (BD Biosciences). The flow cytometry data were analyzed with FlowJo software (Tree Star). The antibody used in this study are listed in **Supplementary Table 1** [↗](#).

4. ILC3s sorting, transfer and co-culture experiment

For ILC3s transfer experiment, co-culture experiment and single-cell RNA-Seq, ILCs were collected by FACS. Small intestine lamina propria cells were prepared and stained with surface molecules for 30 min at 4°C followed by sorting on a FACS Aria III cell sorter (BD Biosciences). A gating strategy of live Lineage[−]CD127⁺KLRG1[−]c-Kit⁺ was used to sort the ILC3s.

Purification checks were performed after each sort. The cells were suspended in phosphate buffered saline (PBS) and then intravenously injected into HFD mice (200 µL PBS/mouse). ILC3s were sorted from 20 WT NCD mice, and sorting for the adoptive transfer was performed six times within 4 weeks from the first injection. The total number of ILC3 injected into HFD mice was 2.4×10^5 cells/mouse/experiment. Mice were placed on an HFD for 16 weeks before ILC3s were transferred. In addition, to confirm that these cells indeed originate from NCD mice, we transferred ILC3s from CD45.1 mice to WT mice and detected the CD45.1 positive cells in recipient mice.

5. Single-cell RNA-Seq

Single cells were captured via the GemCode Single Cell Platform using the GemCode Gel Bead, Chip and Library Kits (10X Genomics) according to the manufacturer's protocol. Briefly, flow-sorted cells were suspended in PBS containing 0.4% BSA and loaded at 6,000 cells per channel. The cells

were then partitioned into a GemCode instrument, where individual cells were lysed and mixed with beads carrying unique barcodes in individual oil droplets. The products were subjected to reverse transcription, emulsion breaking, cDNA amplification, shearing, 5' adaptor and sample index attachment. Libraries were sequenced on a HiSeq 2500 (Illumina).

6. Isolation of stromal vascular fraction cells (SVF) and induction of beigeing

Isolation of SVF and induction of beigeing were performed as reported^[48]. Briefly, subcutaneous white adipose tissue (sWAT) from male C57BL/6 mice was digested with 1 mg/mL type 2 collagenase (Sigma-Aldrich) at 37°C for 40 minutes with shaking. Digestion was terminated by complete DMEM/F12 medium containing 10% fetal bovine serum (FBS), 100 units/mL penicillin and 100 units/mL streptomycin (Invitrogen, CA, USA). The cell suspension was centrifuged at 700×g for 10 minutes to separate floating adipocytes from the SVF pellets. The pellets containing the SVF cells were resuspended in complete medium and filtered using a 70 µm diameter filter. The cell suspension was then centrifuged at 700×g for 10 minutes. Cell pellets were resuspended and mixed well and then plated on dishes in complete medium. Cells were grown to 95% confluence in complete medium and then differentiated into beige or white adipocytes as previously described^[49]. Briefly, for induction to beige adipocytes, cells were cultured for 2 days with induction medium supplemented with 5 µg/mL insulin, 1 nmol/L T3, 1 µmol/L rosiglitazone, 125 µmol/L indomethacin, 0.5 mmol/L isobutylmethylxanthine (IBMX) and 5 µmol/L dexamethasone. Cells were then cultured in maintenance medium supplemented with 5 µg/mL insulin, 1 nmol/L T3 and 1 µmol/L rosiglitazone for 4 days. Fresh media were replaced every 2 days.

7. Co-culture experiments

To determine the direct effect of ILC3s on beige adipocytes differentiation, SVF were co-cultured with ILC3s using a transwell system (0.4 µm pore size, BD Biosciences). In brief, SVF isolated from adipose tissue were grown in the bottom chamber of the transwell insert in a 12-well plate and induced into beige adipocytes, while sorted ILC3s were seeded in the upper chamber. SVFs cultured alone and SVF cells co-cultured with CD127⁺ cells were used as controls. After co-culture for 72 h, beige adipocytes in the lower chamber were collected for further detection.

8. RNA isolation and qPCR analysis

RNA was extracted from cells or adipose tissue using RNeasy (Applied Gene, Beijing, China) and reverse-transcribed into cDNAs with Hifair™ III 1st Strand cDNA Synthesis SuperMix (Yeasten). SYBR Green-based quantitative real-time PCR was performed using the Agilent Aria Mx real-time PCR system. The primer sequences used in this study are listed in **Supplementary Table 2**.

9. Western blot analysis

Differentiated SVF cells and sWAT were homogenized using RIPA lysis buffer. Proteins were separated by 10% sodium dodecyl sulfate–polyacrylamide gel electrophoresis and then transferred onto a nitrocellulose (NC) membrane. Membranes were incubated in 5% fat-free milk for 1 hour at room temperature and then incubated with primary antibodies overnight at 4°C. The reaction was detected with IRDye-conjugated secondary antibody and visualized using the Odyssey infrared imaging system (LI-COR Biosciences).

10. Histological studies

Paraffin-embedded sWAT sections were stained with hematoxylin-eosin (H&E). Images were scanned using a NanoZoomer-SQ (Hamamatsu).

11. Oil red O staining

Cells were washed with PBS for three times, and then fixed with 4% paraformaldehyde for 30 minutes. Next, the cells were stained with oil red O for 60 minutes, and washed with PBS for 5 minutes. Cells were then stained with hematoxylin for 5 seconds and washed with PBS. The dyed cells were photographed under the microscope (Leica, Germany).

12. Enzyme-linked immunosorbent assay

Levels of IL-22 were measured by double-antibody sandwich enzyme-linked immunosorbent assays (ELISAs) (M2200; R&D Systems). Briefly, 100 μ L of Assay Diluent were added to each well. 50 μ L of blood samples and standards were added and incubated at room temperature for 2 hours. After washing, conjugates were added and incubated for 2 hours. After washing, the substrate solution was added and incubated for 30 minutes and washed. And stop solution were added finally. Optical density (OD) values were measured using a microplate reader (Bio-Rad, Hercules, CA, USA).

13. OGTT and ITT

Oral Glucose tolerance tests (OGTT) and insulin tolerance tests (ITT) were performed after 16 or 6 hours of fasting, respectively. Mice were given with 3 g/kg glucose by gavage or injected intraperitoneally with 0.75 U/kg insulin. Blood samples were collected from the tail vein at 0, 15, 30, 60, 90 and 120 min after glucose or insulin treatment. Blood glucose levels were measured using a glucometer (Roche, Basel, CH).

14. Cold exposure

Mice were placed in a 4°C cold room for 6 hours. The rectal temperature was measured every hour during the cold challenge with a rectal probe (Braintree Scientific, Braintree, MA).

15. Statistical analysis

Data were analyzed by GraphPad Prism software v.8.0 and presented as the mean $\pm$ s.e.m. The Shapiro–Wilk normality test was used to determine the normal distribution of samples. Unpaired Student's *t* test (normal distribution) or Mann–Whitney U-tests (non-normal distribution) was used to analyze data between two groups and one-way ANOVA followed by Bonferroni's multiple-comparisons test (normal distribution) or Kruskal–Wallis test (non-normal distribution) was used for three or more groups. The sample sizes were determined by power analysis using StatMate v.2.0. No data were excluded during the data analysis.

Author contributions

Weizhen Zhang, Yue Yin and Hong Chen designed the study; Hong Chen and Lijun Sun performed most of the experiments and analyzed the data; Lu Feng, Xue Han, Yunhua Zhang, Wenbo Zhai and Zehe Zhang helped with the experiments; Hong Chen wrote the original draft; Weizhen Zhang reviewed and edited the manuscript; Yue Yin and Weizhen Zhang obtained funding and supervised the study. All authors have read and agreed to the published version of the manuscript.

Competing interests

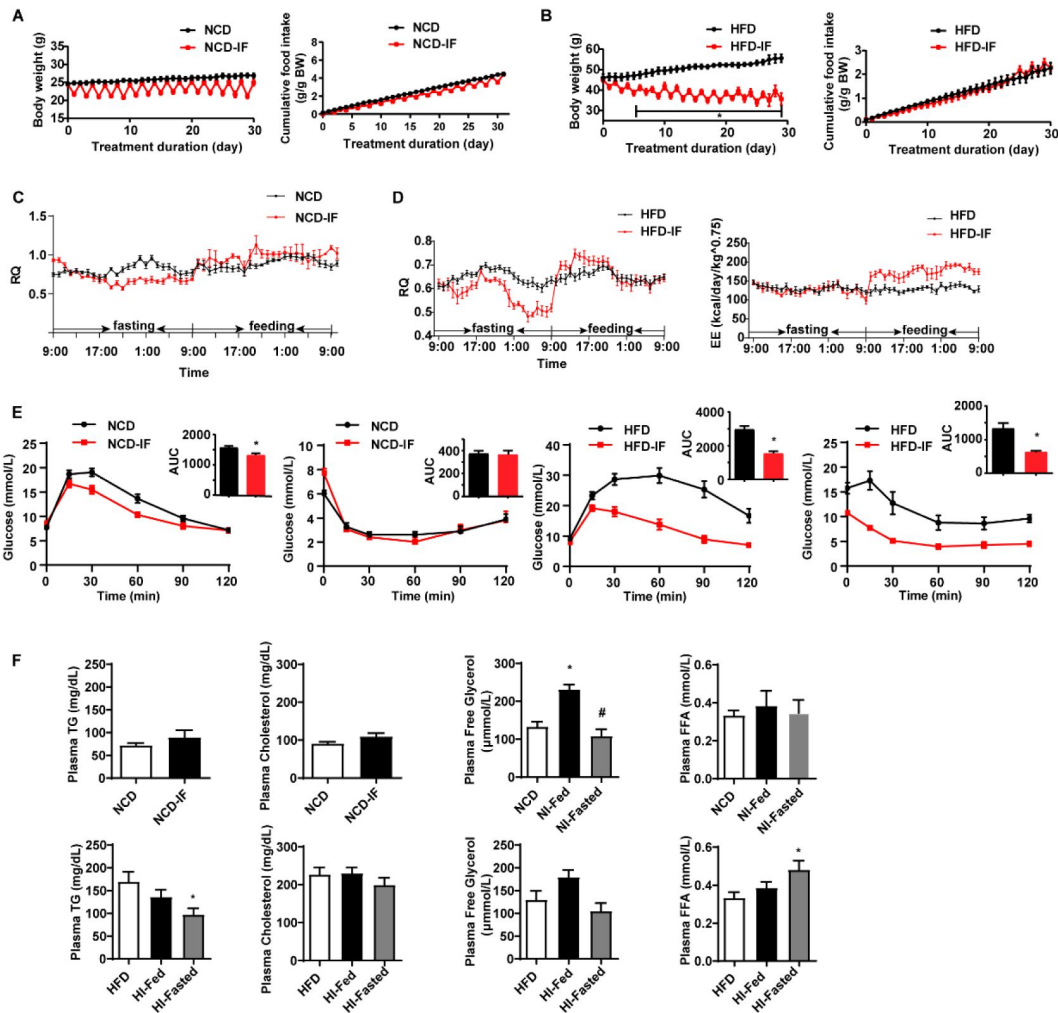
The authors declare no competing interests.

Acknowledgements

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Data availability

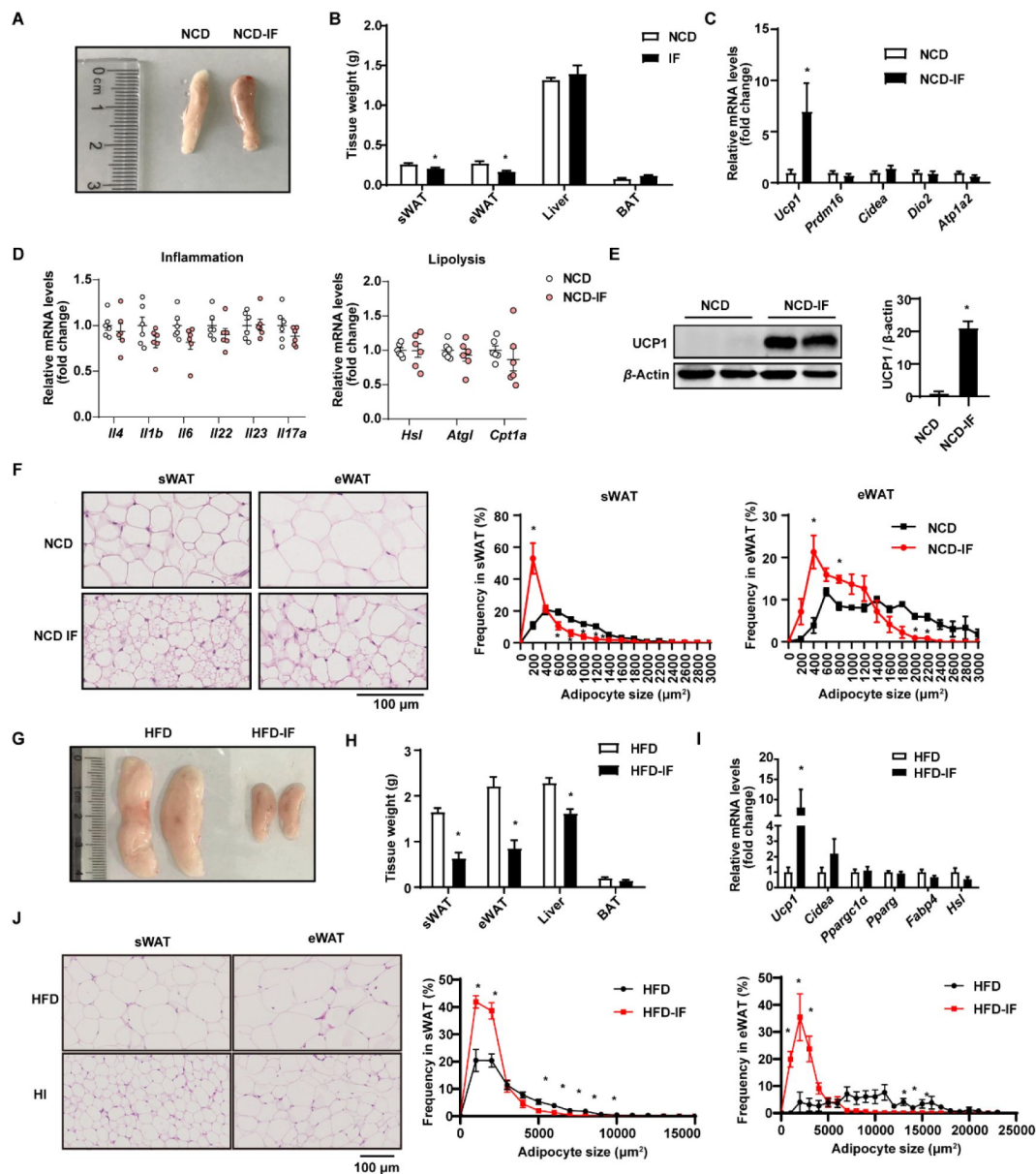
Data that support the findings in this study are available from corresponding authors upon reasonable request.



Supplementary Figure 1

Intermittent fasting improves glucose metabolism and lipid metabolism.

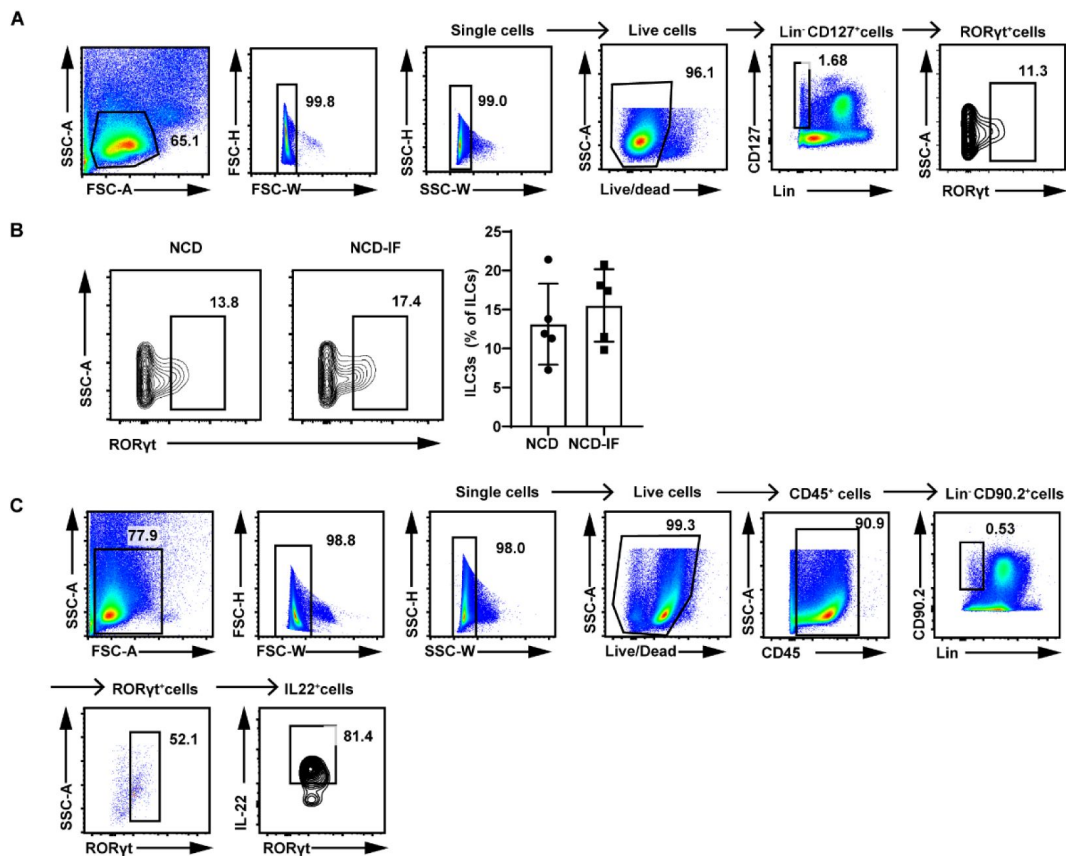
- A. Body weight and cumulative food intake normalized to body weight of mice fed NCD with or without intermittent fasting. n=9.
- B. Body weight and cumulative food intake normalized to body weight of mice fed HFD with or without intermittent fasting. n=9.
- C. Respiratory quotient (RQ) of mice fed NCD with or without intermittent fasting. n=5.
- D. Respiratory quotient (RQ), energy expenditure (EE) of mice fed HFD with or without intermittent fasting. n=5.
- E. Oral glucose tolerance test (OGTT) and area under the curve (AUC). Insulin tolerance test (ITT) and area under the curve (AUC) of NCD, NCD-IF, HFD and HFD-IF mice. n=6 per group.
- F. Levels of plasma triglycerides (TG), cholesterol, free glycerol and free fatty acids (FFA). n=6 mice/group.
- All data represent the mean $\pm$ s.e.m. Statistical significance was determined by unpaired two-tailed Student's *t* test or one-way ANOVA with Bonferroni's multiple-comparisons test.



Supplementary Figure 2

Intermittent fasting promotes white adipose tissue beiging.

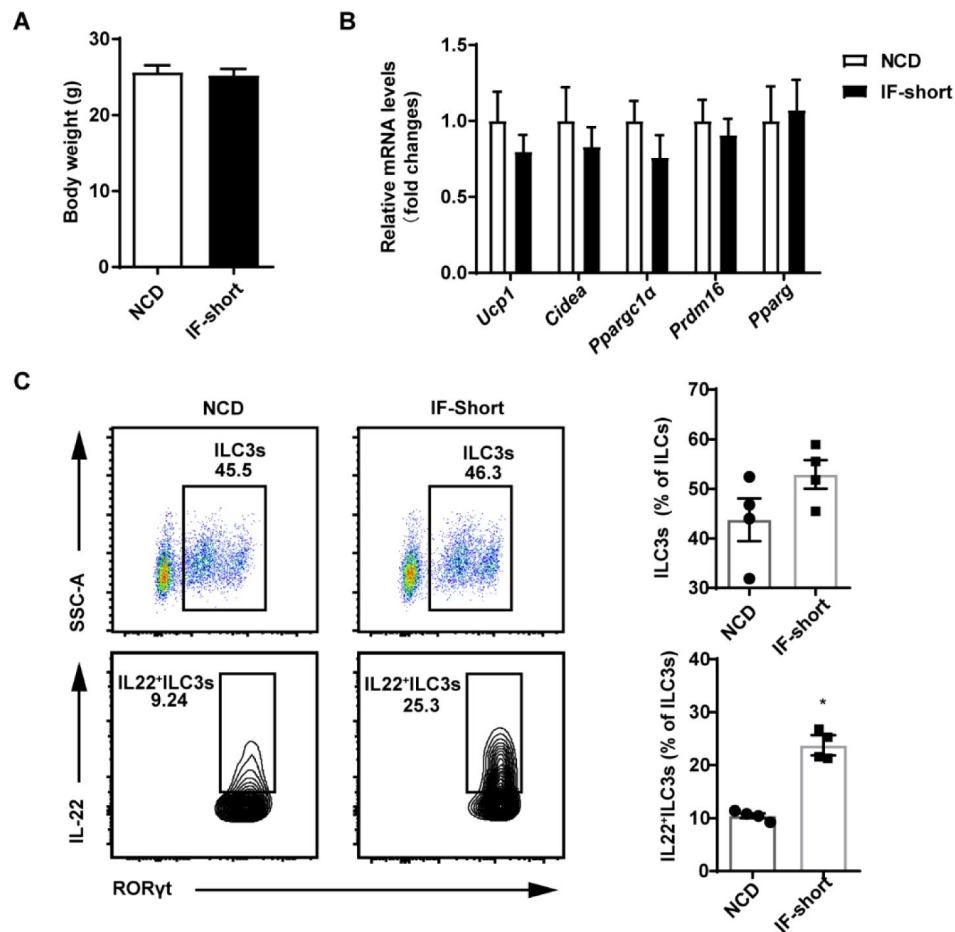
- A. Representative image of sWAT from mice fed NCD with or without intermittent fasting.
- B. Tissue weight of sWAT, eWAT, Liver, BAT, from NCD and NCD-IF mice. $n=6$.
- C. qPCR analysis of thermogenic genes in sWAT from NCD and NCD-IF mice. $n=6$ mice/group.
- D. qPCR analysis of inflammation and lipolysis genes in sWAT from NCD and NCD-IF mice. $n=6$ mice/group.
- E. UCP1 protein expression in the sWAT of NCD and NCD IF mice detected by Western blotting. β -actin was used as the loading control. The relative protein signal intensity was quantified using ImageJ software.
- F. H&E staining in sWAT and eWAT. Cell sizes were measured using ImageJ. $n=5$ per group. Scale bar: 100 μ m.
- G. Representative image for sWAT of HFD and HFD-IF mice. $n=6$.
- H. Tissue weight of sWAT, BAT, Liver, eWAT from HFD and HFD-IF mice. $n=6$.
- I. qPCR analysis of thermogenic genes in sWAT from HFD and HFD-IF mice. $n=6$ mice/group.
- J. H&E staining in sWAT and eWAT from HFD and HFD-IF mice. Cell sizes were measured using ImageJ. $n=5$ per group. Scale bar: 100 μ m.
- All data represent the mean $\pm$ s.e.m. Statistical significance was determined by unpaired two-tailed Student's t test (B-E, H, I).



Supplementary Figure 3

Intermittent fasting demonstrates no effect on the number of ILC3s in sWAT.

- A. Gating strategy for flow cytometry analysis of live lineage[−]CD127⁺RORγt⁺ILC3s in the SVF of sWAT.
 B. Flow cytometric analysis of RORγt⁺ ILC3s in live CD127⁺ lineage[−] ILCs from the sWAT of NCD mice, NCD-IF mice. n=5.
 C. Gating strategy for flow cytometry analysis of IL-22⁺ cells in live CD45⁺ lineage[−]CD90.2⁺RORγt⁺ILC3s in the SVF of sWAT.
 All data represent the mean ± s.e.m. Statistical significance was determined by unpaired two-tailed Student's *t* test.



Supplementary Figure 4

Short-term intermittent fasting induces intestinal ILC3s to secrete IL-22.

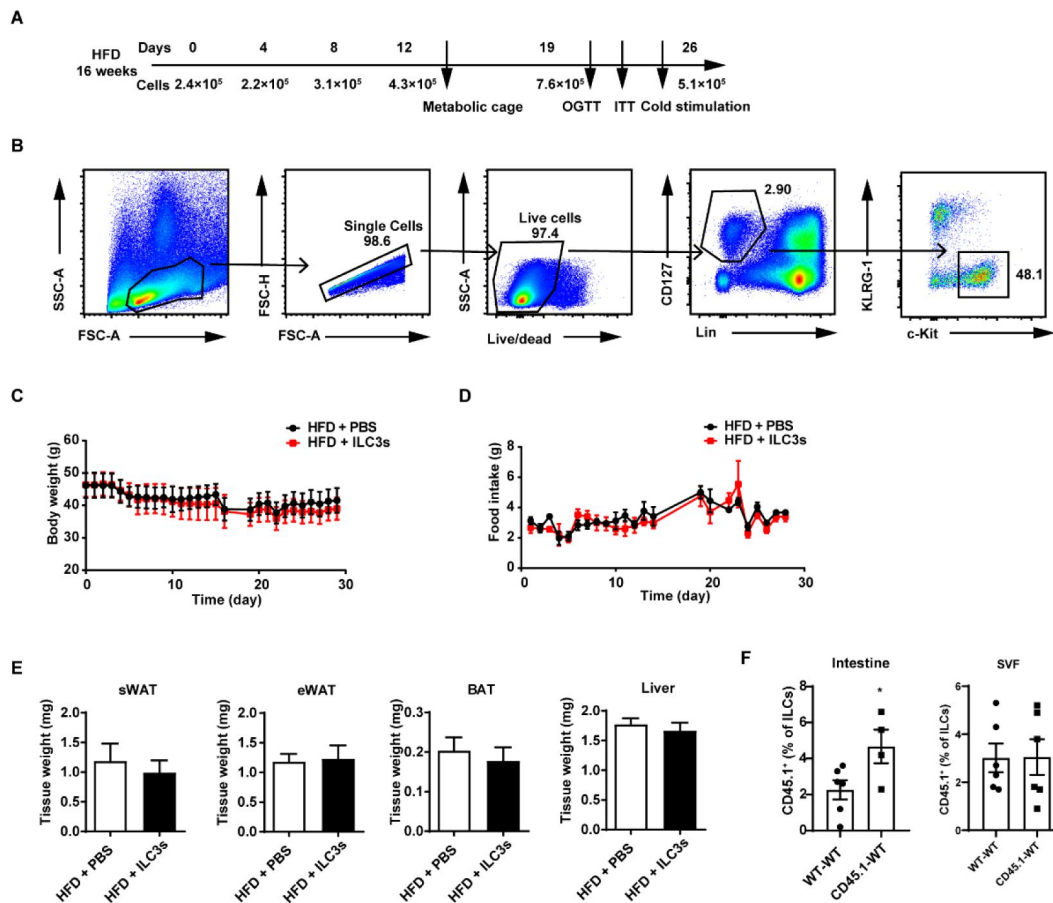
Eight-week-old SPF mice were exposed to one cycle of intermittent fasting for 2 days (IF-short), while the control group (NCD) has free access to NCD. n=6.

A. Body weight of NCD and IF-short mice. n=6.

B. qPCR analysis of thermogenic genes in sWAT from NCD and IF-short mice. n=6 mice/group.

C. Flow cytometric analysis of ROR γ ⁺ ILC3s in live CD127⁺ lineage⁻ ILCs in the small intestine lamina propria of NCD or IF-short mice. Flow cytometric analysis of IL-22⁺ cells in live CD127⁺ lineage⁻ ROR γ ⁺ ILC3s from the small intestine lamina propria of NCD or IF-short mice. Four independent experiments were performed with similar results. n=4.

All data represent the mean $\pm$ s.e.m. Statistical significance was determined by unpaired two-tailed Student's *t* test.



Supplementary Figure 5

Adoptive transfer of ILC3s has no effect on the body weight in mice fed HFD.

A. Timeline for the recipient mice transferred with PBS or ILC3s. Mice were fed with HFD for 16 weeks and then injected with ILC3s intravenously six times in a month.

B. Gating strategy for flow sorting of live CD127⁺ lineage⁻ c-kit⁺ KLRG1⁺ ILC3s in the siLP.

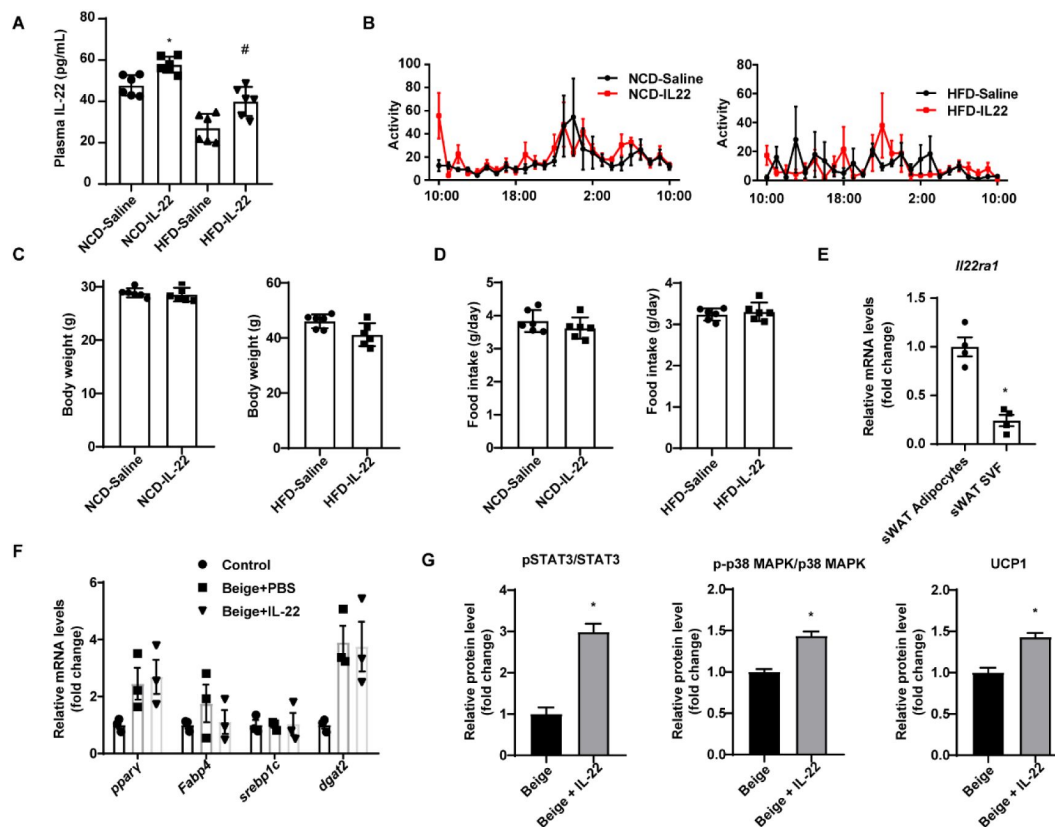
C. Body weight of HFD mice transferred with ILC3s or control. n=6 per group.

D. Food intake of HFD mice transferred with ILC3s or control. n=6 per group.

E. Tissue weight of sWAT, eWAT, BAT, and liver from HFD mice transferred with ILC3s or control. n=6 per group.

F. Transfer of ILC3s from CD45.1 mice to WT mice. The percentage of CD45.1⁺ cells in ILCs in the small intestine and SVF. n=4-6.

All data represent the mean $\pm$ s.e.m. Statistical significance was determined by unpaired two-tailed Student's *t* test.



Supplementary Figure 6

Exogenous IL-22 has no effect on the body weight of mice.

A. Levels of IL-22 in plasma of NCD-Saline, NCD-IL-22, HFD-Saline, HFD-IL-22 mice. n=6 mice/group. # vs HFD-Saline.

B. Activity of mice fed NCD or HFD which were intraperitoneally injected with IL-22 or saline. n=5 for each group.

C. Body weight of NCD and HFD mice intraperitoneally injected with IL-22 or saline. n=6 for each group.

D. Daily food intake of NCD and HFD mice intraperitoneally injected with IL-22 or saline. n=6 for each group.

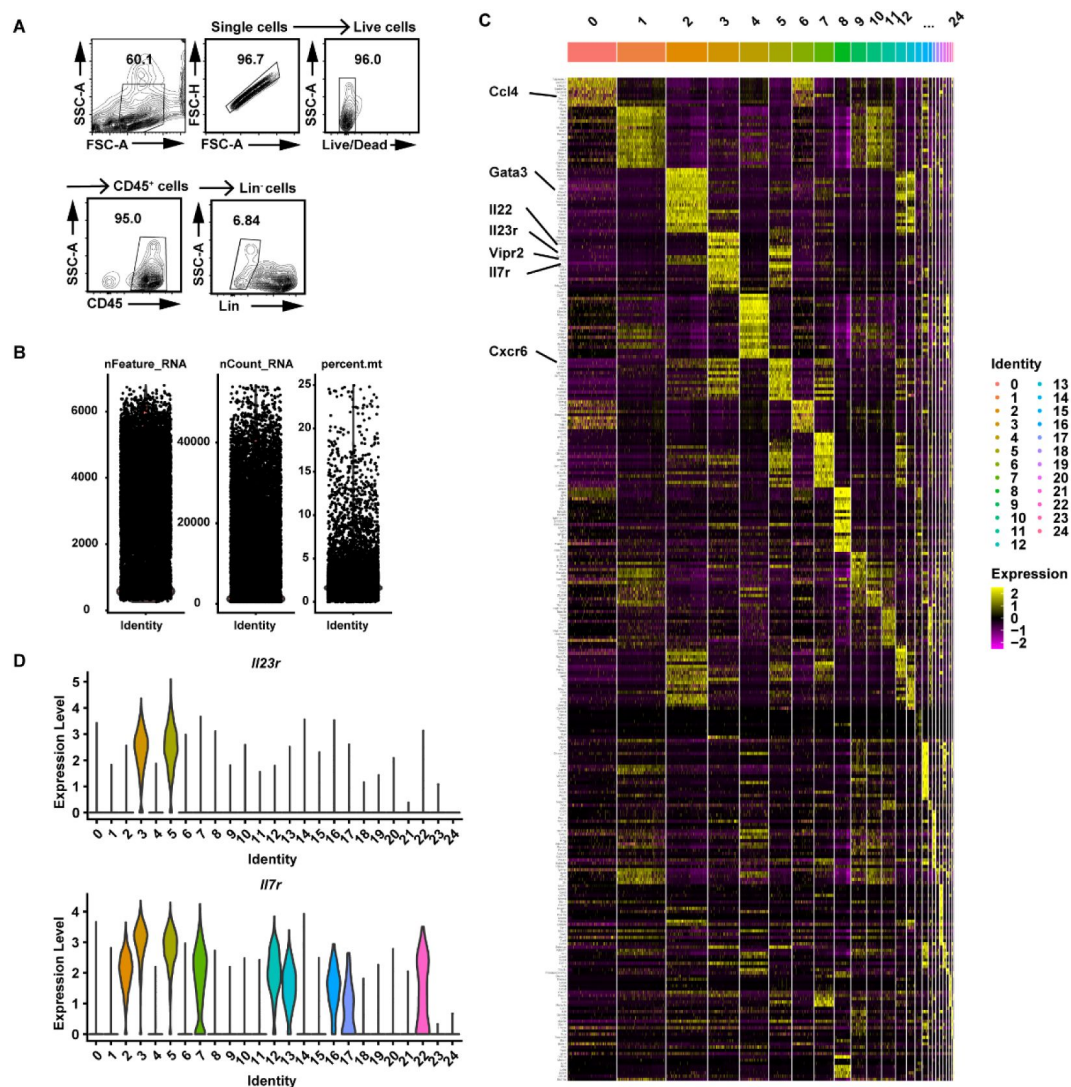
E. qPCR analysis of *Il22ra1* genes in adipocytes or SVF cells isolated from sWAT. n=4.

* denotes $P < 0.05$ vs sWAT adipocytes.

F. qPCR analysis of adipogenic marker genes of SVF cells and adipocytes. n=3.

G. Relative protein signal intensity quantified using Image J software.

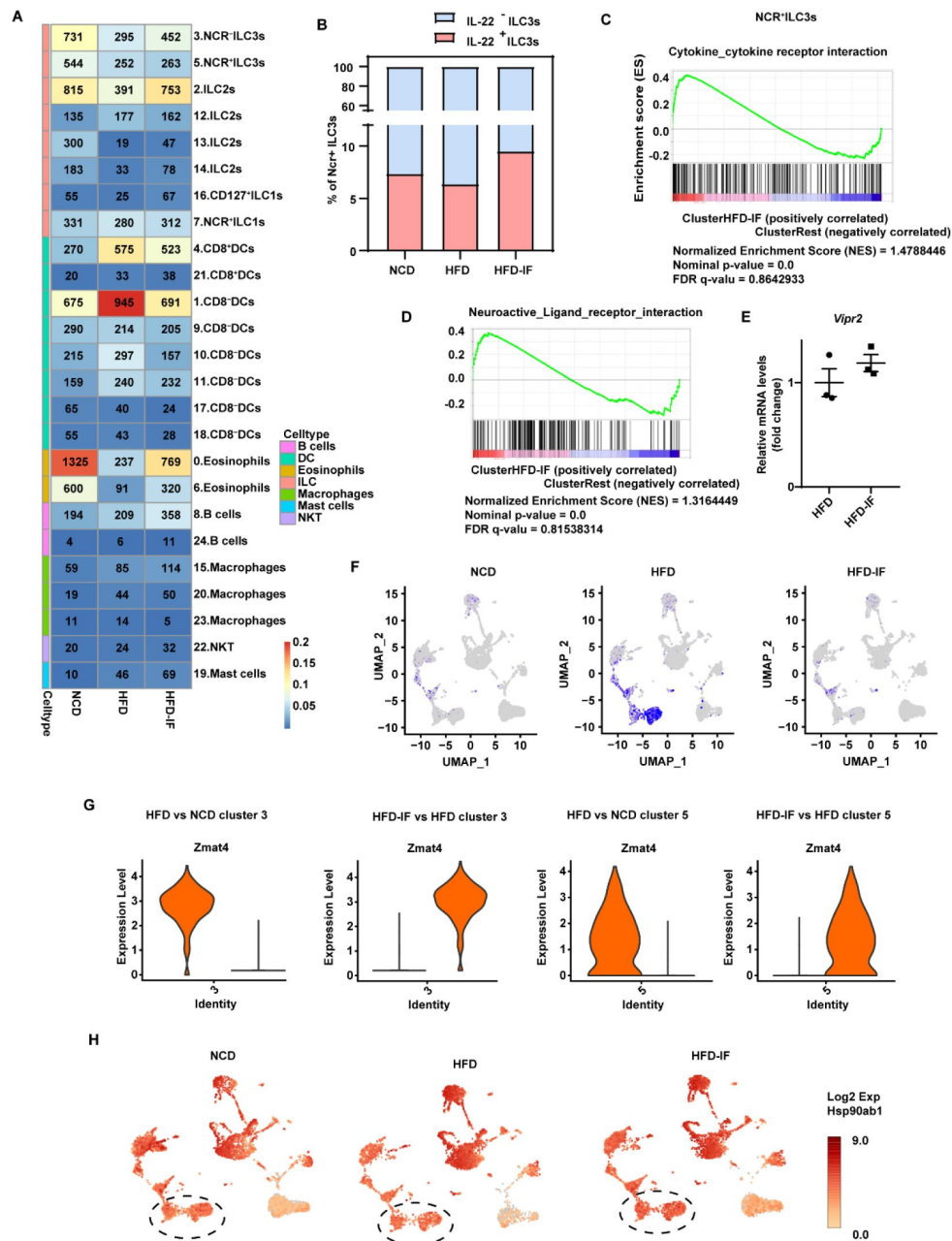
All data represent the mean $\pm$ s.e.m. Statistical significance was determined by one-way ANOVA (A, F) with Bonferroni's multiple-comparisons test or unpaired two-tailed Student's *t* test (B-E,G).



Supplementary Figure 7

Profiling of intestinal immune cells from mice fed NCD, HFD or HFD-IF

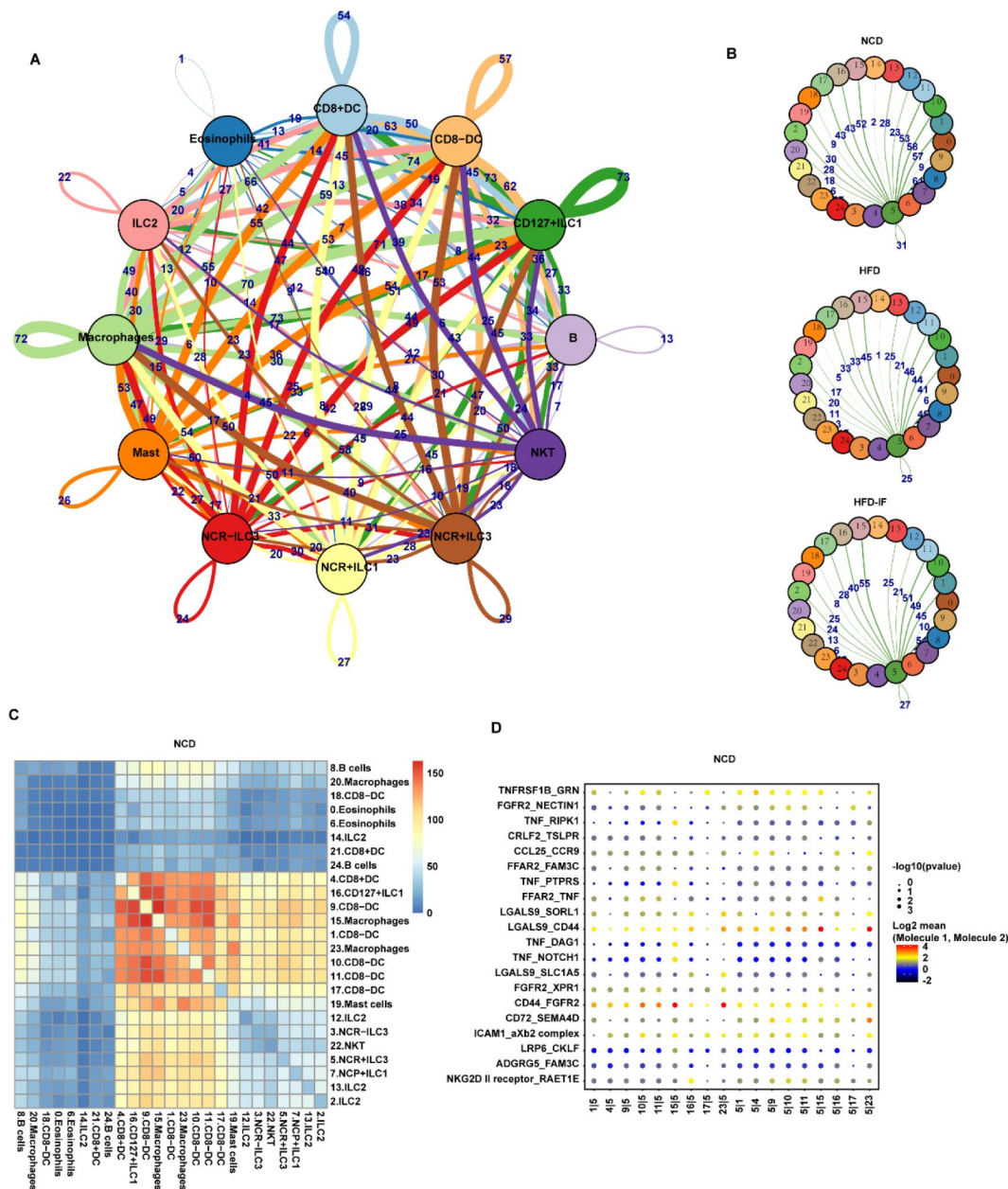
- A. Gating strategy for flow cytometry sorting of Live CD45⁺ Lineage⁻ immune cells in the small intestine lamina propria.
- B. scRNA-seq data quality control of sorted immune cells from mice fed NCD, HFD or HFD-IF.
- C. Unbiased heatmap of the gene levels of the top 20 unique cluster marker genes for each cell cluster. Cluster identities are shown above the heatmap.
- D. Violin plots showing RNA expression of cluster markers for the indicated cell populations. UMAP feature plots are based on the UMAP shown in [Figure 6A](#).



Supplementary Figure 8

Effects of Intermittent fasting on the gene expression of ILC3s.

- A. Number and proportion of cells detected in each group in the study. Shown are Fraction (color bar, out of all cells from that group) and number of cells from each of the 25 cell clusters (rows) in each individual group (columns).
- B. Percentage of IL-22⁺ cells defined by the mRNA levels using Loupe Browser 5.0 of NCR⁺ ILC3s in NCD, HFD and HFD-IF mice.
- C. GSEA demonstrating enrichment of cytokine-cytokine receptor interaction pathway genes upregulated by intermittent fasting (HFD-IF mice) vs. control (HFD mice) in NCR⁺ ILC3s.
- D. GSEA demonstrating enrichment of neuroactive ligand-receptor interaction pathway genes upregulated by intermittent fasting (HFD-IF mice) vs. control (HFD mice) in NCR⁺ ILC3s.
- E. qPCR analysis of *Vipr2* in ILC3s sorted from siLP of HFD or HFD-IF mice.
- F. UMAP feature plots showing RNA expression of *Zmat4* based on the UMAP shown in [Figure 6B](#).
- G. Violin plots showing RNA expression of *Zmat4* in NCR⁺ILC3s (cluster 3) and NCR⁺ILC3s (cluster 5).
- H. UMAP feature plots showing RNA expression of *Hsp90ab1* based on the UMAP shown in [Figure 6B](#).



Supplementary Figure 9

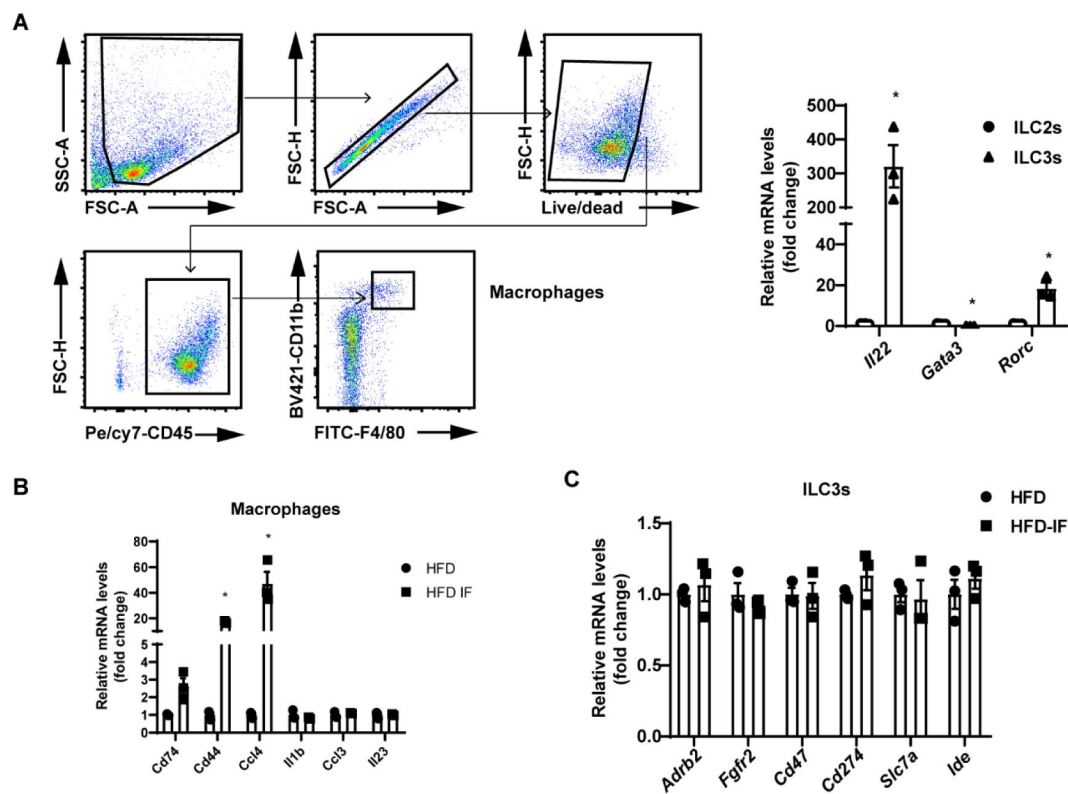
Cell-cell interactions in the intestine.

A. Connectome web analysis of intestine immune cell interacting cell types based on the expression of the ligand in mice fed NCD. The vertex (colored cell node) represents the cell cluster. The thickness of the connecting lines is proportional to the number of interactions between the nodes.

B. Connectome web analysis of NCR⁺ ILC3s with the 25 cell clusters based on expression of the ligand. The vertex (colored cell node) represents the cell cluster, while the thickness of the connecting lines is proportional to the number of interactions between two nodes.

C. Heatmap showing the number of significant interactions identified between cell types in sorted siLP immune cells of NCD mice as determined by CellphoneDB. The color represents the number of interactions between cell types: a higher number of interactions (red) and a lower number of interactions (blue).

D. Interaction pattern of the top 20 protein pairs and the top 20 cell types. The x-axis is the cell type interaction, and the y-axis is the protein interaction. The larger the point is, the smaller the p value. The color represents the average expression. Red to black indicates the level from high to low.



Supplementary Figure 10

Increased expression of CD44 and CCL4 in macrophages.

Flow cytometry sorted macrophages and ILC3s from HFD and HFD-IF mice were used to detect the mRNA levels of proteins involved in the interaction of macrophages and ILC3s.

A. Gating strategy for flow sorting of macrophages in the siLP. qPCR analysis of *Il22*, *Gata3*, and *Rorc* in ILC2s and ILC3s. n=3/group.

B. mRNA levels of proteins involved in the interaction and *Il23* in macrophages. n=3/group.

C. mRNA levels of proteins involved in the interaction in ILC3s. n=3/group.

All data represent the mean $\pm$ s.e.m. Statistical significance was determined by unpaired two-tailed Student's *t* test.

<i>Gene</i>	Forward Primer	Reverse Primer
<i>Il22</i>	ATGAGTTTTTCCCTTATGGGGAC	GCTGGAAGTTGGACACCTCAA
<i>Il17</i>	TCCCTCTGTGATCTGGGAA	CTCGACCCTGAAAGTGAAGG
<i>Il1β</i>	CCTTCCAGGATGAGGACATGA	TGAGTCACAGAGGATGG-GCTC
<i>Il6</i>	TAGTCCTTCCTACCCCAATTTC	TTGGTCCTTAGCCACTCCTTC
<i>Ifnγ</i>	ATGAACGCTACACACTGCATC	CCATCCTTTTGCCAGTTCCTC
<i>Tbp</i>	ACCTTATGCTCAGGGCTTGG	GCCGTAAGGCATCATTGGAC
<i>RpL32</i>	GAGCAACAAGAAAACCAAGCA	TGCACACAAGCCATCTACTCA
<i>Hprt2</i>	TCAGTCAACGGGGGACATAAA	GGGGCTGTACTGCTTAACCAG
<i>Hsl</i>	CCAGCCTGAGGGGCTTACTG	CTCCATTGACTGTGACATCTCG
<i>Adrb3</i>	GGCCCTCTCTAGTTCCCAG	TAGCCATCAAACCTGTTGAGC

Table S1.

Sequences of primers used in quantitative PCR.

<i>Chrebpβ</i>	AGATGGAGAACCGACGTATCA	ACTGAGCGTGCTGACAAGTC
<i>Dio2</i>	AATTATGCCTCGGAGAAGACCG	GGCAGTTGCCTAGTGAAAGGT
<i>Atp1a2</i>	CCACCACTGCGGAAAATGG	GCCCTTAGACAGATCCACTTGG
<i>Atgl</i>	AACACCAGCATCCAGTTCAA	GGTTCAGTAGGCCATTCTC
<i>Fabp4</i>	ACACCGAGATTCCTTCAAACCTG	CCATCTAGGGTTATGATGCTCTTCA
<i>Cidea</i>	ATCACAACTGGCCTGGTTACG	TACTACCCGGTGTCCATTCT
<i>Ppargc1α</i>	GATTGAAGTGGTGTAGCGAC	GTCGCTACACCACTTCAATC
<i>Pparg</i>	TCAGCTCTGTGGACCTCTCC	ACCCTTGCATCCTTCACAAG
<i>Adrb3</i>	GGCCCTCTCTAGTTCCCAG	TAGCCATCAAACCTGTTGAGC
<i>Prdm16</i>	CAGCACGGTGAAGCCATTC	GCGTGCATCCGCTTGTG
<i>Il4</i>	GGTCTCAACCCCCAGCTAGT	GCCGATGATCTCTCTCAAGTGAT
<i>Ucp1</i>	CACCTTCCCGCTGGACACT	CCCTAGGACACCTTTATACCTAATGG
<i>Il23a</i>	ATGCTGGATTGCAGAGCAGTA	ACGGGGCACATTATTTTTAGTCT
<i>Tnf</i>	CGTCGTAGCAAACCACCAAG	GAGATAGCAAATCGGCTGACG
<i>Adrb2</i>	GGGAACGACAGCGACTTCTT	GCCAGGACGATAACCGACAT
<i>Fgfr2</i>	CCTCGATGTCGTTGAACGGTC	CAGCATCCATCTCCGTCACA
<i>Cd47</i>	TGGTGGGAAACTACACTTGCG	CGTGCGGTTTTTCAGCTCTAT
<i>Cd274</i>	GCTCCAAAGGACTTGACGTG	TGATCTGAAGGGCAGCATTTC
<i>Slc7a1</i>	CTGCCTCAACACCTATGACCT	GAGAGCAGCAATCAAGAAGGAG
<i>Ide</i>	AATCCGGCCATCCAGAGAATA	GGGTCTGACAGTGAACCTATGT
<i>Cd74</i>	AGTGCGACGAGAACGGTAAC	CGTTGGGGAACACACACCA
<i>Cd44</i>	TCGATTTGAATGTAACCTGCCG	CAGTCCGGGAGATACTGTAGC

Table S1. (continued)

<i>Ccl4</i>	TTCCTGCTGTTTCTCTTACACCT	CTGTCTGCCTCTTTTGGTCAG
<i>Il1b</i>	GCAACTGTTTCCTGAACTCAACT	ATCTTTTGGGGTCCGTCAACT
<i>Ccl3</i>	TTCTCTGTACCATGACACTCTGC	CGTGGAATCTTCCGGCTGTAG
<i>Gata3</i>	CTCGGCCATTTCGTACATGGAA	GGATACCTCTGCACCGTAGC
<i>Roryt</i>	GACCCACACCTCACAAATTGA	GACCCACACCTCACAAATTGA

Table S1. (continued)

Antibody and clone	Source	cat#
Cell Staining Buffer	BioLegend	420201
eBioscience™ Fixation/Perm Diluent	Invitrogen	00-8333-56
eBioscience™ Fixable Viability Dye eFluor™ 606	Invitrogen	65-0866-14
PerCP/Cy5.5 anti-mouse CD45(30-F11)	BioLegend	103131
FITC anti-mouse CD3ε(RA3-6B2)	BioLegend	103205
FITC anti-mouse/human CD45R/B220(RA3-6B2)	BioLegend	103205
FITC anti-mouse Ly-6G/Ly-6C(Gr-1)(RB6-8C5)	BioLegend	108405
FITC anti-mouse CD19(6D5)	BioLegend	115506
FITC anti-mouse CD5(53-7.3)	BioLegend	100605
BV421 anti-mouse CD127(IL-7Rα) (A7R34)	BioLegend	135023
PE/Cyanine7 anti-mouse CD90.2(30-H12)	BioLegend	105325
PE/Cyanine7 anti-mouse CD45.1	BioLegend	110729
PE anti-mouse RORγt(Q31-378)	BD Pharmingen	562607
Alexa Fluor 647 anti-mouse IL-22(Poly5164)	BioLegend	516406
BV605 anti-mouse/human KLRG1(MAFA)(2F1/KLRG1)	BioLegend	138419
PE anti-mouse CD117(c-kit) (2B8)	BioLegend	105807
Rb polyclonal antibody to UCP1	abcam	ab10983
Phospho-Stat3 (Tyr705) Rabbit mAb	Cell signaling technology	#9145
Stat3 (124H6) Mouse mAb	Cell signaling technology	#9139
Phospho-p38 MAPK (Thr180/Tyr182) (D3F9) XP® Rabbit mAb	Cell signaling technology	#4511
p38 MAPK (D13E1) XP® Rabbit mAb	Cell	#8690
	signaling technology	

Table S2.

Key resources

References

- [1] Liu B, Du Y, Wu Y, Snetselaar L G, Wallace R B, Bao W (2021) **Trends in obesity and adiposity measures by race or ethnicity among adults in the united states 2011-18: Population based study** *Bmj* **372**
- [2] Colman R J *et al.* (2009) **Caloric restriction delays disease onset and mortality in rhesus monkeys** *Science (New York, N.Y.)* **325**:201–204
- [3] Mattison J A *et al.* (2012) **Impact of caloric restriction on health and survival in rhesus monkeys from the nia study** *Nature* **489**:318–321
- [4] Mattison J A, Colman R J, Beasley T M, Allison D B, Kemnitz J W, Roth G S, Ingram D K, Weindruch R, de Cabo R, Anderson R M (2017) **Caloric restriction improves health and survival of rhesus monkeys** *Nature Communications* **8**
- [5] Ulgherait M, Midoun A M, Park S J, Gatto J A, Tener S J, Siewert J, Klickstein N, Canman J C, Ja W W, Shirasu-Hiza M (2021) **Circadian autophagy drives iTRF-mediated longevity** *Nature* **598**:353–358
- [6] Stekovic S *et al.* (2019) **Alternate day fasting improves physiological and molecular markers of aging in healthy, non-obese humans** *Cell Metabolism* **30**:878–881
- [7] de Cabo R, Mattson M P (2019) **Effects of intermittent fasting on health, aging, and disease** *New England Journal of Medicine* **381**:2541–2551
- [8] Hepler C *et al.* (2022) **Time-restricted feeding mitigates obesity through adipocyte thermogenesis** *Science* **378**:276–284
- [9] Mattson M P, Moehl K, Ghena N, Schmaedick M, Cheng A (2018) **Intermittent metabolic switching, neuroplasticity and brain health** *Nature Reviews. Neuroscience* **19**:63–80
- [10] Liu Y *et al.* (2019) **Sirt3 mediates hippocampal synaptic adaptations to intermittent fasting and ameliorates deficits in app mutant mice** *Nature Communications* **10**
- [11] Mattson M P, Arumugam T V (2018) **Hallmarks of brain aging: Adaptive and pathological modification by metabolic states** *Cell Metabolism* **27**:1176–1199
- [12] Kim K-H *et al.* (2017) **Intermittent fasting promotes adipose thermogenesis and metabolic homeostasis via VEGF-mediated alternative activation of macrophage** *Cell Research* **27**:1309–1326
- [13] Li G *et al.* (2017) **Intermittent fasting promotes white adipose browning and decreases obesity by shaping the gut microbiota** *Cell Metabolism* **26**:672–685
- [14] Gasteiger G, Fan X, Dikiy S, Lee S Y, Rudensky A Y (2015) **Tissue residency of innate lymphoid cells in lymphoid and nonlymphoid organs** *Science (New York, N.Y.)* **350**:981–985
- [15] Vivier E *et al.* (2018) **Innate lymphoid cells: 10 years on** *Cell* **174**:1054–1066

- [16] O'Sullivan T E, Rapp M, Fan X, Weizman O-E, Bhardwaj P, Adams N M, Walzer T, Dannenberg A J, Sun J C (2016) **Adipose-resident group 1 innate lymphoid cells promote obesity-associated insulin resistance** *Immunity* **45**:428–441
- [17] Wang H *et al.* (2019) **Adipose group 1 innate lymphoid cells promote adipose tissue fibrosis and diabetes in obesity** *Nat Commun* **10**
- [18] Brestoff J R *et al.* (2015) **Group 2 innate lymphoid cells promote beiging of white adipose tissue and limit obesity** *Nature* **519**:242–246
- [19] Lee M W, Odegaard J I, Mukundan L, Qiu Y, Molofsky A B, Nussbaum J C, Yun K, Locksley R M, Chawla A (2015) **Activated type 2 innate lymphoid cells regulate beige fat biogenesis** *Cell* **160**:74–87
- [20] Wang L *et al.* (2021) **Adiponectin restrains ILC2 activation by ampk-mediated feedback inhibition of IL-33 signaling** *J Exp Med* **218**
- [21] Sanos S L, Bui V L, Mortha A, Oberle K, Heners C, Johnen C, Diefenbach A (2009) **Rorgammat and commensal microflora are required for the differentiation of mucosal interleukin 22-producing NKp46+ cells** *Nature Immunology* **10**:83–91
- [22] Wang X *et al.* (2014) **Interleukin-22 alleviates metabolic disorders and restores mucosal immunity in diabetes** *Nature* **514**:237–241
- [23] Hasnain S Z *et al.* (2014) **Glycemic control in diabetes is restored by therapeutic manipulation of cytokines that regulate beta cell stress** *Nature Medicine* **20**:1417–1426
- [24] Qi X *et al.* (2019) **Gut microbiota–bile acid–interleukin-22 axis orchestrates polycystic ovary syndrome** *Nature Medicine* **25**:1225–1233
- [25] Sasaki T, Moro K, Kubota T, Kubota N, Kato T, Ohno H, Nakae S, Saito H, Koyasu S (2019) **Innate lymphoid cells in the induction of obesity** *Cell Reports* **28**:202–217
- [26] Upadhyay V *et al.* (2012) **Lymphotoxin regulates commensal responses to enable diet-induced obesity** *Nat Immunol* **13**:947–953
- [27] Wang Y, Kuang Z, Yu X, Ruhn K A, Kubo M, Hooper L V (2017) **The intestinal microbiota regulates body composition through nfil3 and the circadian clock** *Science* **357**:912–916
- [28] Kawano Y *et al.* (2022) **Microbiota imbalance induced by dietary sugar disrupts immune-mediated protection from metabolic syndrome** *Cell* **185**:3501–3519
- [29] Seillet C *et al.* (2020) **The neuropeptide vip confers anticipatory mucosal immunity by regulating ILC3 activity** *Nature Immunology* **21**:168–177
- [30] Victor A R *et al.* (2017) **IL-18 drives ILC3 proliferation and promotes IL-22 production via nf-kb** *J Immunol* **199**:2333–2342
- [31] Gronke K *et al.* (2019) **Interleukin-22 protects intestinal stem cells against genotoxic stress** *Nature* **566**:249–253
- [32] Kim M H, Taparowsky E J, Kim C H (2015) **Retinoic acid differentially regulates the migration of innate lymphoid cell subsets to the gut** *Immunity* **43**:107–119

- [33] Mackley E C *et al.* (2015) **Ccr7-dependent trafficking of rory⁺ ilcs creates a unique microenvironment within mucosal draining lymph nodes** *Nat Commun* **6**
- [34] Yu H B *et al.* (2021) **Vasoactive intestinal peptide promotes host defense against enteric pathogens by modulating the recruitment of group 3 innate lymphoid cells** *Proc Natl Acad Sci U S A* **118**
- [35] Zou J, Chassaing B, Singh V, Pellizzon M, Ricci M, Fythe M D, Kumar M V, Gewirtz A T (2018) **Fiber-mediated nourishment of gut microbiota protects against diet-induced obesity by restoring IL-22-mediated colonic health** *Cell Host Microbe* **23**:41–53
- [36] Qi X *et al.* (2019) **Gut microbiota-bile acid-interleukin-22 axis orchestrates polycystic ovary syndrome** *Nat Med* **25**:1225–1233
- [37] Li H, Dong M, Liu W, Gao C, Jia Y, Zhang X, Xiao X, Liu Q, Lin H (2021) **Peripheral IL-6/stat3 signaling promotes beiging of white fat** *Biochim Biophys Acta Mol Cell Res* **1868**
- [38] Zhang Y *et al.* (2016) **Irisin exerts dual effects on browning and adipogenesis of human white adipocytes** *Am J Physiol Endocrinol Metab* **311**:E530–541
- [39] Talbot J, Hahn P, Kroehling L, Nguyen H, Li D, Littman D R (2020) **Feeding-dependent vip neuron-ILC3 circuit regulates the intestinal barrier** *Nature* **579**:575–580
- [40] Yu H B *et al.* (2021) **Vasoactive intestinal peptide promotes host defense against enteric pathogens by modulating the recruitment of group 3 innate lymphoid cells** *Proceedings of the National Academy of Sciences of the United States of America* **118**
- [41] Stockinger B, Di Meglio P, Gialitakis M, Duarte J H (2014) **The aryl hydrocarbon receptor: Multitasking in the immune system** *Annu Rev Immunol* **32**:403–432
- [42] Ibiza S *et al.* (2016) **Glial-cell-derived neuroregulators control type 3 innate lymphoid cells and gut defence** *Nature* **535**:440–443
- [43] Chen H, Sun L, Feng L, Mulholland M, Zhang W, Yin Y (2022) **Peptidoglycan inhibits beiging of adipose tissue** *Acta Pharm Sin B* **12**:990–993
- [44] Hildreth A D, Ma F, Wong Y Y, Sun R, Pellegrini M, O’Sullivan T E (2021) **Single-cell sequencing of human white adipose tissue identifies new cell states in health and obesity** *Nature Immunology* **22**:639–653
- [45] Hu B *et al.* (2020) **γδ T cells and adipocyte il-17rc control fat innervation and thermogenesis** *Nature* **578**
- [46] Wang Q *et al.* (2021) **Il-27 signalling promotes adipocyte thermogenesis and energy expenditure** *Nature* **600**:314–318
- [47] Brooks J F, Behrendt C L, Ruhn K A, Lee S, Raj P, Takahashi J S, Hooper L V (2021) **The microbiota coordinates diurnal rhythms in innate immunity with the circadian clock** *Cell* **184**:4154–4167
- [48] Aune U L, Ruiz L, Kajimura S (2013) **Isolation and differentiation of stromal vascular cells to beige/brite cells** *J Vis Exp* **73**

- [49] Seale P, Conroe H M, Estall J, Kajimura S, Frontini A, Ishibashi J, Cohen P, Cinti S, Spiegelman B M (2011) **Prdm16 determines the thermogenic program of subcutaneous white adipose tissue in mice** *The Journal of Clinical Investigation* **121**:96–105

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Editors

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Reviewer #1 (Public Review):

In the present study, the authors carefully evaluated the metabolic effects of intermittent fasting on normal chow and HFD fed mice and reported that intermittent fasting induces beiging of subcutaneous white adipose tissue. By employing complementary mouse models, the authors provided compelling evidence to support a mechanism through ILC3/IL-22/IL22R pathway. They further performed comprehensive single-cell sequencing analyses of intestinal immune cells from lean, obese, obese undergone intermittent fasting mice and revealed altered interactome in intestinal myeloid cells and ILC3s by intermittent fasting via activating AhR. Overall, this is a very interesting and timely study uncovering a novel connection between intestine and adipose tissue in the context of executing metabolic benefits of intermittent fasting.

(1) The authors showed increased plasma IL-22 and its expression in intestine. Are intestinal ILC3s the main source of plasma IL-22?

(2) The authors transplanted intestinal ILC3s from NCD mice to DIO mice and showed significant metabolic improvements. However, in Fig. 1, intermittent fasting increased IL-22-positive ILC3s proportion rather than changing the total number. Please clarify whether this transplantation is due to increasing ILC3s number or introducing more IL-22 positive ILC3s (which are decreased in DIO). Are these transplanted ILC3s by default homing to intestine rather than to other tissues?

(3) The authors adopted cold challenge at 4 degree for 6 hours to assess beiging in subcutaneous WAT and showed difference in core temperature. However, thermogenesis in this acute cold challenge is mainly by brown adipose tissue. Beiging is a chronic and adaptive response. Based on the data in WAT, there is a beiging phenotype, but the core body temperature in acute cold challenge is not an accurate readout. It would be a missed opportunity by not evaluating thermogenic activity in BAT.

More browning genes should be included to strengthen the beiging phenotype of WAT. Moreover, inflammation in WAT can be examined to provide a whole picture of adipose tissue remodeling through this pathway.

(4) For the SVF beige adipocyte differentiation, 100 ng/mL IL-22 was used. This is highly above the physiological concentration at ~5 pg/mL. Please justify this high concentration used.

The authors showed increased Ucp1 and Cidea expression by IL-22 treatment in SVFs. Please be aware that these increases are likely due to boosted adipogenesis as told by the morphology. Please examine more adipogenic markers to confirm. Is this higher adipogenesis caused by the high concentration of IL-22?

In line 201, the authors drew the conclusion that IL-22 increased SVF beige differentiation. To fully support this conclusion, the authors should assure adipogenesis at the same baseline and then compare beiging, or examine the effect of IL-22 on normal adipogenesis to compare with beige differentiation.

<https://doi.org/10.7554/eLife.91060.2.sa2>

Reviewer #2 (Public Review):

Summary:

This study aims to investigate the mediatory role of intestinal ILC3-derived IL-22 in intermittent fasting-elicited metabolic benefits.

Strengths:

The observation of induction of IL-22 production by intestinal ILC3 is significant, and the scRNAseq provides new information into intestine-resident immune cell profiling in response to repeated fasting and refeeding.

Weaknesses:

The experimental design for some studies needs to be improved to enhance the rigor of overall study. There is a lack of direct evidence showing that the metabolically beneficial effects of IF are mediated by intestinal ILC3 and their derived IL-22. The mechanism by which IL-22 induces thermogenic program is unknown. The browning effect induced by IF may involve constitutive activation of lipolysis, which was not considered.

Majority of weaknesses have been addressed in the revision. Based on the analysis of thermogenic genes in addition to Ucp1 (Fig. 4D and S6F), the alteration on thermogenesis induced by IL-22 is dependent on UCP1 but not other markers such as PGC1a, PPAR α , and Cidea. The data need to be discussed in the Section of Discussion.

<https://doi.org/10.7554/eLife.91060.2.sa1>

Reviewer #3 (Public Review):

Chen et al. investigated how intermittent fasting causes metabolic benefits in obese mice and find that intestinal ILC3 and IL-22-IL-22R signaling contribute to the beiging of white adipose tissue (WAT) and consequent metabolic benefits including improved glucose and lipid

metabolism in diet-induced obese mice. They demonstrate that intermittent fasting causes increased IL22+ILC3 in small intestines of mice. Adoptive transfer of purified intestinal ILC3 or administration of exogenous IL-22 can lead to increases in UCP1 gene expression and energy expenditure as well as improved glucose metabolism. Importantly, the above metabolic benefits caused by intermittent fasting are abolished in IL-22R^{-/-} mice. Using an in vitro experiment, the authors show that ILC3-derived IL-22 may directly act on adipocytes to promote SVF beige differentiation. Finally, by performing sc-RNA-seq analysis of intestinal immune cells from mice with different treatments, the authors indicate a possible way of intestinal ILC3 being activated by intermittent fasting. Overall, this study provides a new mechanistic explanation for the metabolic benefits of intermittent fasting and reveals the role of intestinal ILC3 in the enhancement of the whole-body energy expenditure and glucose metabolism likely via IL-22-induced beige adipogenesis.

Although this study presents some interesting findings, particularly IL-22 derived from intestinal ILC3 could induce beiging of WAT by directly acting on adipocytes, the experimental data are not sufficient to support the key claims in the manuscript.

<https://doi.org/10.7554/eLife.91060.2.sa0>

Author Response

The following is the authors' response to the original reviews.

Comment 1: The authors showed increased plasma IL-22 and its expression in the intestine. Are intestinal ILC3s the main source of plasma IL-22?

Reply: ILC3s are the main source of IL-22 as reported previously (PMID: 30700914). In the small intestine, ILC3s account for about 62% of IL22+ cells. Other IL22+ cells include $\gamma\delta$ T, Foxp3+T and CD4+T cells.

Comment 2: The authors transplanted intestinal ILC3s from NCD mice to DIO mice and showed significant metabolic improvements. However, in Fig. 1, intermittent fasting increased IL-22positive ILC3s proportion rather than changing the total number. Please clarify whether this transplantation is due to increasing ILC3s number or introducing more IL-22 positive ILC3s (which are decreased in DIO). Are these transplanted ILC3s by default homing to the intestine rather than to other tissues?

Reply: We believe that the transplantation increases ILC3s number, leading to the increment in IL22 levels. The transplanted ILC3s by default are homing to the intestine rather than to other tissues because ILC3s express several homing receptors such as CCR7, CCR9, and $\alpha 4\beta 7$, which modulate their capacity to migrate to the gut (PMID: 26141583; PMID: 26708278; PMID: 25575242; PMID: 34625492). Our observation that ILC3s in adipose tissue remained unchanged by ILC3 cell transplantation (Supplementary Figure 5F) also supports this concept.

Comment 3: Thermogenesis in this acute cold challenge is mainly by brown adipose tissue. Beiging is a chronic and adaptive response. Based on the data in WAT, there is a beiging phenotype, but the core body temperature in acute cold challenge is not an accurate readout. It would be a missed opportunity by not evaluating thermogenic activity in BAT. More browning genes should be included to strengthen the beiging phenotype of WAT. Moreover, inflammation in WAT can be examined to provide a whole picture of adipose tissue remodeling through this pathway.

Reply: Per suggestion, we performed additional experiments to measure levels of inflammation genes such as Il4, Il1b, Il6, Il22, Il23, Il17a. As shown in supplemental figure 2D,

these inflammation relevant genes were not altered.

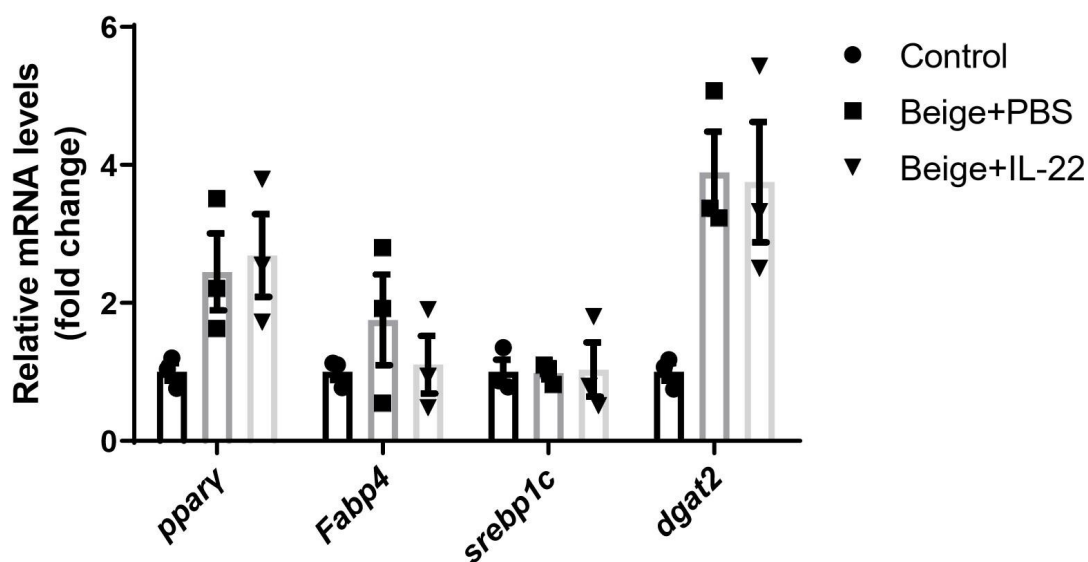
Comment 4: For the SVF beige adipocyte differentiation, 100 ng/mL IL-22 was used. This is highly above the physiological concentration at ~5 pg/mL. Please justify this high concentration used.

Reply: We agree with the reviewer that the dose of IL-22 used is high. However, the efficient dose at 100 ng/ml used in our studies is consistent with the literatures. Previous reports have shown that IL-22 directly activates Stat3 in adipose tissue and primary adipocytes, and promotes the expression of genes involved in triglyceride lipolysis (Lipe and Pnpla2) and fatty-acid β -oxidation (Acox1) at the dose of 100 ng/ml (Wang X, Ota N, et al. Nature. 2014). Consistently, other studies have reported that IL-22 at 100 ng/ml significantly reversed the enhanced expression of CCL2, CCL20 and IL1B mRNAs in granulosa cells in vitro (Qi X, et al. Nat Med. 2019).

Comment 5: The authors showed increased Ucp1 and Cidea expression by IL-22 treatment in SVFs. Please be aware that these increases are likely due to boosted adipogenesis as told by the morphology. Please examine more adipogenic markers to confirm. Is this higher adipogenesis caused by the high concentration of IL-22?

Reply: Per suggestion, we examined the expression of adipogenic marker genes such as Ppar γ and Fabp4. We found that IL-22 did not increase the levels of these adipogenic marker genes relevant to the PBS control as shown in supplemental figure 6F.

Author response image 1.



Comment 6: In line 201, the authors drew the conclusion that IL-22 increased SVF beige differentiation. To fully support this conclusion, the authors should assure adipogenesis at the same baseline and then compare beigeing, or examine the effect of IL-22 on normal adipogenesis to compare with beige differentiation.

Reply: We examined the expression of adipogenic marker genes such as Ppar γ and Fabp4 and found that IL-22 did not increase the expression of these adipogenic marker genes relevant to the PBS control.

Reviewer #2:

This study aims to investigate the mediatory role of intestinal ILC3-derived IL-22 in intermittent fasting-elicited metabolic benefits.

Strengths:

The observation of induction of IL-22 production by intestinal ILC3 is significant, and the scRNAseq provides new information into intestine-resident immune cell profiling in response to repeated fasting and refeeding.

Weaknesses:

The experimental design for some studies needs to be improved to enhance the rigor of the overall study. There is a lack of direct evidence showing that the metabolically beneficial effects of IF are mediated by intestinal ILC3 and their derived IL-22. The mechanism by which IL-22 induces a thermogenic program is unknown. The browning effect induced by IF may involve constitutive activation of lipolysis, which was not considered.

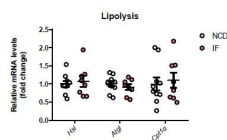
Comment 1: Lack of direct evidence showing that IL-22-expressing ILC3s in intestine is the key contributor to intermittent fasting (IF)-mediated elevation of circulating IL-22 levels. The fraction of IL-22-expressing cells was increased threefold by IF but the increase in circulating IL-22 is moderate (Figs. 1J and 1K).

Reply: IL-22 in circulation is subjected to clearance, degradation, and binding with plasma proteins, et al. Thus, circulating levels of IL-22 may be much lower than the amount secreted by the intestinal IL-22 positive ILC3s.

Comment 2: The loss of fat mass by IF suggests that the active lipolysis may explain the white fat browning which was not considered. This may apply to the observations in IL-22 treated mice as well as IL-22R KO mice.

Reply: We analyzed the expression of genes relate to lipolysis in NCD and NCD-IF mice and found that IF did not alter the levels of these genes in white adipose tissues (Supplementary figure 2D). We have addressed this concerns in lines 119, page 6.

Author response image 2.



Comment 3: IL-22 administration and adoptive transfer of ILC3 had no significant effect on body weight. Not clear how IL-22 improves insulin sensitivity in this case.

Reply: Our results are consistent with previous report showing that IL-22 administration improves insulin sensitivity without change in body weight (Qi X, et al. Nat Med. 2019). In addition, previous studies have demonstrated that IL-22 can increase Akt phosphorylation in muscle, liver and adipose tissues, leading to improvement in insulin sensitivity (Wang X, et al. Nature. 2014). We have addressed this potential mechanism in lines 192-195, page 9.

Comment 4: The energy expenditure data look unusual given that there was little increase in oxygen consumption during dark cycle compared to light cycle (Fig.3).

Reply: The not so obvious difference in oxygen consumption between dark cycle and light cycle may be due to the technical problem of the system.

Comment 5: The thermogenic capacity for the whole fat pad needs to consider the expression of UCP1 in certain amount of tissue and the total mass for each individual animal because the mRNA level itself does not reflect the whole tissue capacity.

Reply: We used the whole subcutaneous adipose tissue from one side for qPCR to reflect the whole tissue capacity.

Comment 6: The design of studies for the adoptive transfer of ILC3 was concerned. The PBS is not a good control for the group with ILC3 cells (Figs. 2A-2H). Similar issue applies for the co-culture study in which beige only is not an ideal control for Beige+ILC3 (Figs. 2I-2J).

Reply: We agree with the reviewer that the PBS is not a good control. Because we cannot find a similar immune cell without any effect on adipocytes, we designed this experiment based on other studies in which saline or PBS are used as ILC transfer experiment controls (Sasaki T, et al. Cell Rep. 2019; Wang H, et al. Nat Commun. 2019)

Comment 7: The induction of thermogenesis by IL-22 treatment may be related to enhanced differentiation rather than direct activation of thermogenic genes (Figs. 4G and 4H).

Reply: Our observation that IL-22 did not alter the levels of genes related to adipogenesis (Supplemental figure 6F) indicates that IL-22 may not alter the differentiation of adipocytes. We addressed this concern in Lines 211-212, page 10.

Reviewer #3:

Chen et al. investigated how intermittent fasting causes metabolic benefits in obese mice and found that intestinal ILC3 and IL-22-IL-22R signaling contribute to the beiging of white adipose tissue (WAT) and consequent metabolic benefits including improved glucose and lipid metabolism in diet-induced obese mice. They demonstrate that intermittent fasting causes increased IL22+ILC3 in small intestines of mice. Adoptive transfer of purified intestinal ILC3 or administration of exogenous IL-22 can lead to increases in UCP1 gene expression and energy expenditure as well as improved glucose metabolism. Importantly, the above metabolic benefits caused by intermittent fasting are abolished in IL-22R-/- mice. Using an in vitro experiment, the authors show that ILC3-derived IL-22 may directly act on adipocytes to promote SVF beige differentiation. Finally, by performing sc-RNA-seq analysis of intestinal immune cells from mice with different treatments, the authors indicate a possible way of intestinal ILC3 being activated by intermittent fasting. Overall, this study provides a new mechanistic explanation for the metabolic benefits of intermittent fasting and reveals the role of intestinal ILC3 in the enhancement of the whole-body energy expenditure and glucose metabolism likely via IL-22-induced beige adipogenesis.

Although this study presents some interesting findings, particularly IL-22 derived from intestinal ILC3 could induce beiging of WAT by directly acting on adipocytes, the experimental data are not sufficient to support the key claims in the manuscript.

Comment 1: Only increased UCP1 expression on mRNA level is not enough to support the being of WAT. More methods such as western blotting and immunostaining of UCP1 in WAT are needed to confirm the enhanced beige adipogenesis.

Reply: Additional experiments have been performed to measure the UCP1 protein by Western blot. The data is included in Figure 4I and Supplementary Figure 2E.

Comment 2: IL-22 is known to modulate metabolic pathways via multiple downstream functions. The use of whole-body knockout of IL-22R could not exclude the indirect effect on the promotion of being of WAT. Specific deletion of IL-22R in adipose tissues is therefore needed to confirm the direct effect of IL-22 on adipocytes which is suggested by the in vitro study.

Reply: We agreed with the reviewer that specific deletion of IL-22R in adipose tissues is critical to confirm the direct effect of IL-22 on adipocytes. We will generate the AdioQ-IL-22R^{-/-} mice to test this concept further in vivo.

Comment 3: The authors failed to show the cellular distribution of IL-22R in adipose tissues. This is important because the mechanism that explains the increased beige adipogenesis could be different based on the expression of IL-22R in adipose progenitor cells or mature adipocytes. So it is not appropriate to conclude that "IL-22 then directly activates IL-22R on adipocytes, leading to subsequent induction of being of white adipose tissue" in line 407. Additionally, Oil red O staining is needed for Fig 4G and Fig 5J, and protein levels of UCP1 and adipogenesis-related markers are needed to evaluate beige fat differentiation and the whole adipogenesis.

Reply: Per suggestion, we have added the expression of IL-22R in adipose progenitor cells or mature adipocytes (Supplementary Figure 6E). In addition, protein levels of UCP1 and adipogenesis-related markers to evaluate the whole adipogenesis (Figure 4I, Supplementary figure 6F) are now included. We have also addressed this issue in lines 207-215, page 10.

Comment 4: Although the authors provided some hypothesis about how intermittent fasting increases IL-22+ILC3 in small intestines by sc-RNA-seq analysis, some functional assays are needed to identify the factors, for example, how about the levels of macrophage-derived IL-23 or AHR ligands in small intestines and whether they contribute to increased percentages of intestinal IL-22+ILC3 following intermittent fasting.

Reply: We used flow cytometry sorting of macrophages combined with qPCR experiments to preliminarily demonstrate that intermittent fasting increases the expression of molecules such as Cd44 and Ccl4 (Supplementary Figure 10B), which may contribute to the increase in the proportion of IL-22+ ILC3s in the intestine under intermittent fasting. Our observation that IL-23 mRNA levels were not changed indicates that this molecule may not be the major contributor for the communication between macrophage and ILC3s. Other potential molecules such as AHR ligands remain to be explored.

Comment 5: What are the differences between adipose ILC3 and intestinal ILC3? Why do transferred ILC3 only migrate to the small intestine but not WAT of recipient mice? It would be better to examine or at least discuss whether other factors from intestinal ILC3 may also contribute to being of WAT following intermittent fasting.

Reply: Intestinal ILC3s specifically express gut homing receptors CCR7, CCR9, and $\alpha\beta 7$ (PMID: 26141583; PMID: 26708278; PMID: 25575242; PMID: 34625492). This may explain transplantation of intestinal ILC3s can migrate mainly to the intestine instead of adipose

tissue (PMID: 34625492). The proportion of ILC3s in adipose tissue of mice is very small. Their functions have not been clarified yet. We have addressed this issue in lines 156-158, page 8.

There are some other factors from intestinal ILC3 which may also contribute to beiging of WAT following intermittent fasting. By secreting IL-22, ILC3 enhanced the intestinal mucosal barrier, leading to reduction of the influx of LPS and PGN into the bloodstream under high-fat diet conditions, and subsequent increase in the beiging of white adipose tissue (Chen H, et al. *Acta Pharm Sin B*. 2022). We have addressed this potential mechanism in lines 344-347, page 16.

Comment 6: The sensitivity of the IL-22 ELISA kit used in the manuscript was 8.2 pg/mL, according to the information from the methods, however, in Fig. 1J and Fig. 2B, the IL-22 levels in mouse plasma were lower than 6 pg/mL, which was below the sensitivity of the ELISA kit and also the assay range. Please explain.

Reply: We have double-checked the original data and found that we have made a mistake in calculating the concentration of IL-22. We have corrected this error (Fig. 1J, Fig. 2B).

Comment 7: In Fig 7A, the significance of the Hypothesis testing should be marked. In Fig 7F and 7G, the contrast between the two groups is not apparent, other comparing ways could be used to enhance the readability.

Reply: Per suggestion, we have marked the significance of the hypothesis testing between HFD vs NCD and HFD-IF vs HFD in Fig7A. Shown in Fig 7F and 7G are the top 20 enriched interacting proteins between different cell types. The dot plot displays the average expression level and significance of protein interactions in cell types.

Comment 8: The total food intake of fasting mice fed with NCD or HFD was less than those without fasting, and the food intake rate the author showed in Fig S1 represents the value that was normalized to body weight. So the author should describe it precisely In line 114.

Reply: We have revised the statement accordingly in line 114-115.

Comment 9: Western blotting analysis has been described in methods, however, there is no corresponding experimental data in the result part.

Reply: The Western blotting results are now included.